

NTRU-CLSC: Efficient Quantum-Resistant NTRU Lattice-Based Certificateless Signcryption Scheme for VANETs

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Abstract—With the development of the Internet of Things (IoT), ensuring secure, real-time communication and privacy in Vehicular Ad Hoc Networks (VANETs) has become a critical challenge. Certificateless signcryption offers a promising solution by eliminating certificate management issues, however, existing schemes struggle to balance post-quantum security with computational efficiency. Most rely on computationally expensive bilinear pairings or general lattices and often lack robust defenses against public key replacement attacks. To address these issues, we propose NTRU-CLSC, an efficient NTRU Lattice-based certificateless signcryption scheme. We leverage efficient polynomial-ring operations to reduce computational cost. We further adopt the Antrag trapdoor mechanism to bind user public keys to the KGC-generated primary trapdoor, thereby preventing public key replacement attacks. In addition, the scheme supports batch verification to reduce latency in high-concurrency scenarios. Security proofs and performance evaluations demonstrate that the proposed scheme can satisfy the security and privacy requirements of VANETs, while achieving better efficiency compared with existing quantum-resistant signcryption schemes.

Index Terms—Certificateless signcryption, NTRU lattice, VANETs, privacy-preserving.

I. INTRODUCTION

AS THE core backbone of Intelligent Transportation Systems (ITS), VANETs realize key functions such as safety early warning, traffic scheduling, and path optimization through real-time data exchanges—both vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I), which significantly enhances driving safety and traffic operation efficiency [1], [2]. The core communication scenarios of VANETs rely on open wireless channels, sensitive data such as speed and position collected by vehicles via On-Board Units (OBUs) needs to be forwarded to

the Application Servers (ASs) through Road Side Units (RSUs), and this process faces multiple security threats including data tampering, identity forgery, and privacy leakage [3], [4], [5]. In V2V communication, tampered warnings may mislead nearby vehicles. In V2I communication, forged identities may submit false traffic reports, while message exposure may compromise trajectory privacy. Meanwhile, the rapid development of quantum computing technology puts cryptographic systems based on traditional hard problems such as discrete logarithms and large integer factorization at risk of being compromised [6], [7], [8], [9]. Moreover, the characteristics of OBUs in VANETs, such as limited computing resources and high communication frequency, impose stringent requirements on the efficiency of security schemes [1], [10].

Signcryption technology achieves both digital signature and encryption functions in a single operation, balancing the authenticity and confidentiality of data. Compared to the traditional “sign-then-encrypt” combination scheme, it is more suitable for the resource-constrained scenarios of VANETs [11], [12]. Certificateless Public Key Cryptography (CL-PKC) eliminates the need for a certificate management authority and avoids the key escrow issue inherent in identity-based cryptosystems. By co-generating keys with users via the Key Generation Center (KGC), it serves as an ideal framework for VANETs security schemes [13], [14]. In high-concurrency VANETs scenarios, the AS often faces a verification bottleneck when processing massive amounts of data reported by vehicles. Batch verification technology allows the AS to verify multiple signcrypted texts simultaneously, significantly reducing verification latency and computational overhead compared to verifying them one by one [10], [15]. Therefore, the Certificateless Signcryption (CLSC) scheme with batch verification capabilities has emerged as a core technical approach to balancing security and efficiency in VANETs.

Existing security schemes for VANETs still have notable limitations: First, most schemes rely on Elliptic Curve Cryptography (ECC) or bilinear pairings, whose security is based on discrete-logarithm-related hard problems that can be solved in polynomial time by Shor’s algorithm on quantum computers. For example, the certificateless aggregate signature scheme for VANETs [1] optimizes verification efficiency, but it relies on the discrete logarithm problem and thus cannot withstand quantum attacks. Second, existing certificateless schemes based on NTRU lattices primarily provide identity authentication without

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supporting confidentiality. For instance, the NTRU lattice-based certificateless signature scheme for VANETs [3] has quantum resistance, but it only supports identity authentication and fails to meet the requirements for data confidentiality and efficient transmission. Third, existing schemes still face practical deployment limitations in VANETs. Schemes relying on bilinear pairings [16], [17] involve many computationally expensive operations, such as bilinear pairings and exponentiations, and their signcryption delay for a single message often fails to meet the real-time requirement of VANETs. Schemes [18], [19] adopt a fixed-key mechanism and do not account for the long online duration and high communication frequency of VANET nodes. Under long-term observation, attackers may gradually accumulate key-related information, which increases the risk of cumulative attacks and threatens long-term communication security. In addition, although some existing post-quantum schemes [20], [21] are built on lattice-based frameworks, they rely on general lattices and require complex operations such as lattice basis generation, resulting in high computational overhead and making them less suitable for lightweight and low-power OBUs. As a core branch of post-quantum cryptography, NTRU lattice cryptography is secure based on the quantum hardness of the Shortest Vector Problem (SVP) on lattices. Moreover, its operations only involve modular multiplication over polynomial rings and convolution operations, so its computational efficiency is significantly superior to both traditional bilinear pairing-based schemes and general lattice-based schemes.

To address the above limitations, we propose an NTRU lattice-based certificateless signcryption scheme with batch verification for VANETs. The scheme integrates the Antrag trapdoor mechanism [22], dynamic key update, and batch verification to provide efficient signcryption, resistance to public key replacement and cumulative attacks, and scalable verification under high-concurrency traffic. Moreover, it achieves unforgeability and confidentiality in the random oracle model (ROM), resists replay and quantum attacks, and maintains low computational and communication overhead for resource-constrained OBUs.

A. Our Contribution

We summarize the threefold contribution to this work below.

- We propose NTRU-CLSC, an NTRU lattice-based post-quantum certificateless signcryption scheme tailored for VANETs. It provides efficient confidentiality, authentication, and privacy protection for safety-critical V2V and V2I messages. To the best of our knowledge, this is the first VANET-oriented NTRU lattice-based certificateless signcryption scheme that combines post-quantum security with batch verification.
- To address key replacement, long-term key exposure, and high-concurrency verification bottlenecks, the proposed scheme integrates the Antrag trapdoor mechanism, a dynamic key update mechanism, and batch verification. In addition, with pseudonym technology and RSU-side invalid ciphertext filtering, the scheme supports conditional privacy preservation, malicious-vehicle tracing, and more efficient batch processing at the AS.

- Under the ROM, we prove that the scheme satisfies indistinguishability under adaptive chosen-ciphertext attack (IND-CCA2) and unforgeability, with its security properties reduced to the hardness of the Ring Learning With Errors (R-LWE) and Ring Small Integer Solution (R-SIS) problems over NTRU lattices. Moreover, extensive experimental verification and comparative analysis reveal that our scheme significantly reduces the computational overhead and achieves superior efficiency compared to state-of-the-art lattice-based signcryption schemes [21], [23], [24], [25], [26], [27].

B. Outline of This Paper

The paper is organized as follows: Sections II and III present related works and preliminaries. Section IV details the NTRU-CLSC construction. Sections V and VI provide security proofs and performance evaluations, followed by the conclusion in Section VII.

II. RELATED WORKS

Recent representative studies [28], [29], [30] indicate that VANET security schemes should be assessed not only by security and privacy guarantees, but also by computation and communication overhead, key-management complexity, and scalability in dynamic vehicular environments. Guided by these observations, we review existing VANET security schemes from two perspectives, namely traditional cryptographic schemes and post-quantum schemes. As shown in Table I, our discussion focuses on the limitations of prior work in terms of quantum resistance, confidentiality support, deployment efficiency, resistance to public key replacement, dynamic key update, and batch-verification scalability.

A. Traditional Authentication and Signcryption Schemes in VANETs

Before the emergence of post-quantum cryptography, VANET security mainly relied on traditional public-key techniques, which can be broadly categorized into PKI-based, identity-based, and certificateless frameworks. To improve efficiency, Zheng et al. [11] proposed signcryption technology to combine digital signatures and encryption into a single logical step. This primitive offers significantly lower computational costs compared to the traditional sign-then-encrypt approach.

In the developmental timeline of vehicular network security, Malone et al. [31] pioneered identity-based signcryption to alleviate the burden of certificate management. To mitigate the inherent key escrow issue, Barbosa et al. [12] subsequently introduced the inaugural certificateless signcryption protocol utilizing bilinear pairings. Although Liu et al. [32] later constructed a scheme within the standard model, it was identified to be vulnerable to public key replacement attacks. Conversely, the protocol designed by Miao et al. [33] successfully fulfilled the criteria for unforgeability and confidentiality. Targeting edge IoT environments, Zhang et al. [34] engineered an efficient on-line/offline signcryption primitive. More recently, Xie et al. [35]

TABLE I
COMPARATIVE SUMMARY OF REPRESENTATIVE RELATED WORKS

References	Techniques	Advantages	Limitations
Traditional authentication and signcryption schemes			
[12, 32, 33]	Bilinear-pairing-based signcryption	Supporting authentication, confidentiality, and formal analysis	Failing to support post-quantum security and lightweight deployment
[35]	ECC-based signcryption	Supporting authentication and confidentiality	Failing to support post-quantum security, batch verification, and dynamic key update
[28]	ECC-based authentication	Supporting batch verification and scalability	Lack of confidentiality, dynamic key update, and post-quantum security
[29]	Hash function, blockchain, and zero-knowledge proof	Supporting anonymity, privacy preservation, and secure information sharing	Failing to support post-quantum security, batch verification, and dynamic key update
[30]	Hash function, ElGamal, and Bloom filter	Supporting authentication, malicious-vehicle detection, and trust enhancement	Lack of confidentiality, batch verification, and post-quantum security
Post-quantum authentication and signcryption schemes			
[38], [39]	NTRU-based signature	Supporting post-quantum authentication and formal analysis	Lack of confidentiality, batch verification, and dynamic key update
[40, 41, 42]	NTRU-based signature	Supporting authentication, integrity, and aggregation	Failing to support confidentiality, batch verification, and dynamic key update
[43]	SIS-based aggregate signature	Supporting post-quantum authentication and aggregation	Lack of confidentiality, certificateless signcryption, and dynamic key update
[44]	Hash function and lattice-based authentication	Supporting post-quantum authentication, anonymity, and batch verification	Lack of confidentiality, certificateless signcryption, and dynamic key update
[6]	LWE/SIS-based signcryption	Supporting authentication, confidentiality, and formal analysis	Lack of lightweight deployment, batch verification, and dynamic key update
[45]	NTRU-based signcryption	Supporting authentication and confidentiality	Failing to support certificateless design, batch verification, and dynamic key update
[46]	NTRU-based signcryption	Supporting authentication and confidentiality	Lack of lightweight deployment, batch verification, and dynamic key update

applied certificateless signcryption to guarantee secure authentication and transmission in IoV contexts. In parallel, recent traditional VANET security studies have further improved privacy-preserving authentication efficiency. For example, Zhang et al. [36] proposed a hardware-secure-module-based lightweight conditional privacy-preserving authentication scheme, while Yan et al. [28] introduced an edge-assisted hierarchical batch authentication framework to enhance verification scalability. Nevertheless, these conventional frameworks are grounded in the hardness of discrete logarithms or large integer factorization. Consequently, the advent of quantum computing poses a severe threat, as Shor's algorithm [37] could potentially solve these underlying mathematical problems in polynomial time.

B. Post-Quantum Authentication and Signcryption Schemes in VANETs

To resist quantum computing attacks, the academic community has turned to post-quantum cryptography (PQC). Among various PQC candidates, lattice-based cryptography, particularly the NTRU lattice [38], has attracted significant attention due to its reliance on polynomial ring operations, which offer superior efficiency over general lattices. Initial efforts in lattice-based VANET security focused primarily on identity authentication. Tian et al. [39] proposed an early lattice-based certificateless signature (CLS) scheme. Addressing efficiency issues, Xie et al. [40] designed an efficient NTRU-based certificateless signature scheme secure under the ROM. Similarly, Xu et al. [41] and Chen et al. [42] proposed NTRU-based

schemes for medical cyber-physical systems and multi-signature scenarios, respectively. Additionally, Xie et al. [43] introduced a sequential aggregate signature scheme using trapdoor sampling. More recent lattice-based studies have continued to strengthen authentication efficiency and privacy preservation in VANETs. For example, Prajapat et al. [44] proposed a secure lattice-based aggregate signature scheme for VANETs, and Wani et al. [45] designed a dynamic anonymous quantum-secure batch-verifiable authentication scheme. These studies further confirm the suitability of lattice-based techniques for post-quantum vehicular authentication, but they still mainly focus on authentication rather than integrated confidentiality. While these NTRU-based signature schemes provide integrity and authentication, they inherently lack data confidentiality. Using separate signature and encryption schemes introduces excessive overhead incompatible with VANETs. In practice, this separation increases both per-message computation and transmitted bytes, thereby raising channel load and end-to-end latency in high-frequency VANET messaging.

To achieve both confidentiality and authentication efficiently in the quantum era, post-quantum signcryption has become a critical research direction. Yu et al. [6] proposed lattice-based certificateless signcryption schemes resilient to quantum attacks. However, most existing solutions still face limitations regarding efficiency or applicability. For instance, Yan et al. [46] proposed an NTRU-based signcryption scheme, but it adopts a traditional Certificate Authority (CA) framework with heavy management overhead. Yu et al. [47] designed a proxy signcryption scheme based on NTRU lattices. However, its proxy

authorization mechanism introduces redundant complexity ill-suited for VANETs. Furthermore, the absence of simultaneous support for batch verification and robust defense mechanisms against public key replacement attacks in existing post-quantum schemes results in significant verification bottlenecks during high-concurrency scenarios and leaves the system vulnerable to key tampering.

In summary, the above literature indicates that existing VANET security schemes still suffer from three main limitations. Traditional ECC and pairing-based schemes cannot provide post-quantum security. Existing NTRU/lattice-based certificateless schemes mainly support authentication, but do not offer integrated confidentiality protection. Although several post-quantum signcryption schemes have been proposed, they still have difficulty simultaneously achieving lightweight VANET-oriented deployment, resistance to public key replacement, dynamic key update against cumulative attacks, and scalable batch verification. These limitations motivate the design of a VANET-oriented NTRU lattice-based certificateless signcryption scheme that jointly considers post-quantum security, integrated confidentiality and authentication, lightweight deployment, and scalable verification.

III. PRELIMINARIES

This section outlines the preliminaries necessary for the proposed NTRU-CLSC scheme, including notations, mathematical definitions, and underlying hardness assumptions. Furthermore, it describes the system architecture, threat model, and the essential security requirements for VANETs.

A. Notations

Let $\mathbb{Z}_q$ denote the finite field of integers modulo a prime q . The dimension of the polynomial ring is represented by a positive integer p . The polynomial ring itself is defined as $R_q = \mathbb{Z}_q[x]/(x^p + 1)$, which comprises all polynomials with coefficients in $\mathbb{Z}_q$ and degree strictly less than p . The subset of invertible elements within R_q is indicated by $R_q^\times$. Regarding the Gaussian sampling parameters, $\sigma \in \mathbb{R}^+$ denotes the standard deviation of the discrete Gaussian distribution. The parameter α specifies the target quality for trapdoor generation. Additionally, the real numbers r and R (where $0 < r < R$) correspond to the inner and outer radii of the annular sampling domain, respectively.

B. Mathematical Definitions

Definition 1 (NTRU Lattice): Let $q > 5$ be a prime number, a positive integer p denote the ring dimension (corresponding to the polynomial degree), and $a, b \in R_q$ where a is invertible modulo (q) . Let $\mathbf{h} = b * a^{-1} \pmod{q}$. The NTRU lattice is defined as $\mathcal{A}_{\mathbf{h},q} = \{(u, v) \in R^{2p} \mid u + v * \mathbf{h} = 0 \pmod{q}\}$, which is a full-rank lattice on R^{2p} generated by the matrix as shown in (1).

$$\mathcal{A}_{\mathbf{h},q} = \begin{pmatrix} -C_p(\mathbf{h}) & \mathbf{I}_p \\ q\mathbf{I}_p & \mathbf{O}_p \end{pmatrix} \quad (1)$$

where $\mathbf{I}_p$ is a $p \times p$ unit matrix, $\mathbf{O}_p$ is a $p \times p$ null matrix, and $C_p(\mathbf{h})$ is an anti-circulate matrix.

Algorithm 1: Antrag Trapdoor Generation($d, q, r, R, \alpha, \sigma$).

Input: The degree d , the modulus q , a target quality α , and starting radii r, R such that $\sqrt{q}/\alpha < r < R < \alpha\sqrt{q}$.
Output: Trapdoor $\mathbf{B}$, public key $\mathbf{h}$

- 1: **repeat**
- 2: **for** $1 \leq i \leq d/2$ **do**
- 3: uniform annulus sample $(z_i, \omega_i) \in \mathbb{C}^2$ uniformly such that $(|z_i|, |\omega_i|) \in A^+(r, R)$
- 4: **end for**
- 5: $a' \leftarrow \varphi^{-1}(Z_1, \dots, Z_{d/2})$
- 6: $b' \leftarrow \varphi^{-1}(W_1, \dots, W_{d/2})$
- 7: $a \leftarrow \lfloor a' \rfloor$
- 8: $b \leftarrow \lfloor b' \rfloor$
- 9: **until** $(|\varphi_i(a)|, |\varphi_i(b)|) \in A^+(\sqrt{q}/\alpha, \alpha\sqrt{q})$ for all $i = 1, \dots, d/2$
- 10: using the **Hermite Normal Form Algorithm** [48], find $M_1, N_1 \in R$, such that $a \cdot N_1 - b \cdot M_1 = I_p$, Scale this solution by q to obtain: $M_q = q \cdot M_1$, $N_q = q \cdot N_1$
- 11: apply **Babai's nearest plane Algorithm** [49] to find the lattice vector (M, N) closest to (M_q, N_q) within the set of linear combinations: $\{(a, b), (x \cdot a, x \cdot b), \dots, (x^{p-1} \cdot a, x^{p-1} \cdot b)\}$
- 12: Compute the euclidean norm of (M, N) , if $\|(M, N)\| > p \cdot \sigma$, re-execute **Algorithm 1**
- 13: Output the private trapdoor $\mathbf{B} = \begin{pmatrix} a & b \\ M & N \end{pmatrix} \in \mathbb{Z}_q^{2p \times 2p}$ and public key $\mathbf{h} = b \cdot a^{-1} \in R_q^\times$

Algorithm 2: Gaussian Sampling Algorithm.

Input: $\mathbf{B}, \sigma, c$
Output: v

- 1: $v_p \leftarrow 0, c_p \leftarrow c$
- 2: **for** $i \leftarrow n, \dots, 1$ **do**
- 3: $c'_i \leftarrow \frac{\langle \mathbf{c}_i, \mathbf{b}_i \rangle}{\|\mathbf{b}_i\|^2}$
- 4: $a'_i \leftarrow \frac{\sigma}{\|\mathbf{b}_i\|^2}$
- 5: $z_i \leftarrow \text{SampleZ}(a'_i, c'_i)$
- 6: $\mathbf{c}_{i-1} \leftarrow \mathbf{c}_i - z_i \mathbf{b}_i, \mathbf{v}_{i-1} \leftarrow z_i \mathbf{b}_i$
- 7: **end for**
- 8: **return** (z, z')

Definition 2 (Antrag Trapdoor Generation Algorithm): The Antrag trapdoor sampling algorithm based on NTRU [22] is defined as shown in Algorithm 1. Compared with generic GPV/MP-type trapdoors, Antrag is better matched to our NTRU setting because it outputs $\mathbf{h}$ and a compact short basis $\mathbf{B}$ directly, enabling more efficient Gaussian sampling for partial private-key generation.

Definition 3 (Gaussian Sampling Algorithm): Gaussian sampling serves as a fundamental trapdoor mechanism that facilitates sampling from a discrete Gaussian distribution without leaking information regarding the secret short basis [50]. In this work, we adopt the NTRU-adapted Gaussian sampling algorithm, denoted as $\text{GaussianSampler}(\mathbf{B}, \sigma, c)$, as described in [40].

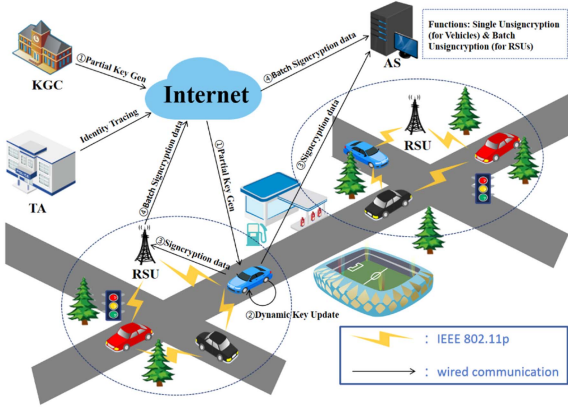


Fig. 1. System model.

In this formulation, B represents the basis of an n -dimensional lattice Λ , while $\tilde{B} = (\tilde{b}_i)_{i \in [n]}$ denotes its Gram-Schmidt orthogonalization. The parameter $\sigma > 0$ indicates the standard deviation, $c \in \mathbb{Z}^n$ specifies the center of the distribution, and the output vector v is drawn from the discrete Gaussian distribution $D_{\Lambda, \sigma, c}$.

C. Hardness Assumption

Definition 4 (NTRU – based SIS Problem): Let $\rho \in \mathbb{Z}^+$, $\beta \in \mathbb{R}$, and $h \in R_{q, \rho}^n$ be the system parameters defining the NTRU lattice $\Lambda_{h, q}$. The Ring-SIS problem over this lattice, denoted as $R\text{-SIS}_{q, 2, \beta}$, requires finding a non-zero vector pair $z = (z_1, z_2) \in \Lambda_{h, q}$ such that the euclidean norm satisfies $0 < \|z\| \leq \beta$.

It is noteworthy that $R\text{-SIS}_{q, 2, \beta}$ constitutes a specific instance of the general Ring-SIS problem. As established by Stehlé et al. [51], there exists a reduction from the $R\text{-SIS}_{q, 2, \beta}$ problem to the γ -Ideal-SVP. Consequently, this problem is widely regarded as computationally intractable.

D. System Model

The system architecture is depicted in Fig. 1, which is composed of five distinct entities. The overall system architecture designates the KGC as an honest-but-curious entity. It honestly executes system initialization and partial private key generation but cannot access the vehicles' full private keys, thereby eliminating key escrow inherent in ID-based systems. The TA is a trusted entity responsible for identity tracing only upon authorization. The AS is a trusted entity that strictly executes unsigncryption and verification logic. Vehicles and RSUs are semi-trusted entities, a vehicle's OBU is tamper-resistant but vulnerable to physical side-channel leakage of secret values, while an RSU performs ciphertext filtration but may process a small fraction of invalid data. The entities achieve secure data transmission through the IEEE 802.11p wireless communication protocol and wired communication. The definitions and functions of the entities are detailed as follows.

- **KGC:** As the core entity of the system, it manages global parameters and supports privacy infrastructure. It generates the system primary key and public parameters using the Antrag trapdoor algorithm, assigns anonymous pseudonyms and partial private keys to vehicles and AS, and maintains an identity-pseudonym mapping database—only granting query access to the TA to enable tracing functions.
- **Vehicle:** As a mobile data sender in VANETs, it is equipped with OBU for communication and cryptographic operations. It registers with KGC to obtain pseudonyms and partial private keys, generates complete public-private key pairs and updates them periodically, performs signcryption on messages before sending them to RSU or AS, and verifies the validity of received partial private keys.
- **AS:** As the data receiving and security verification core of the system, it generates complete public-private key pairs and participates in dynamic key updates. It executes unsigncryption for single messages and supports batch verification of multiple signcrypted texts to reduce computational latency.
- **RSU:** As fixed infrastructure in VANETs, it facilitates data collection. It filters invalid signcrypted texts (via timestamp and norm constraint verification), packages valid ones into a batch, and forwards them to the AS to optimize network transmission.
- **TA:** As a key entity for conditional privacy protection, it is responsible for tracing the identity of malicious vehicles. It receives tracing requests from the AS regarding malicious signcrypted texts, queries KGC's identity-pseudonym mapping database to retrieve the real identity of malicious vehicles, and records tracing results to support accountability.

E. Threat Model

The security properties of the proposed NTRU-CLSC are formally analyzed through game-based interactions with two distinct categories of adversaries.

- **Type-I Adversary:** Adversary $\mathcal{A}_1$ does not hold the system primary key and cannot generate partial private keys for vehicles. However, it can launch public key replacement attacks against the public keys of vehicles or AS, and can obtain public or transmitted data via hash oracle queries and signcryption oracle queries. The only limitation is that it cannot obtain the complete private keys of vehicles.
- **Type-II Adversary:** Adversary $\mathcal{A}_2$ holds the system primary key and can generate partial private keys for vehicles or AS. However, it cannot replace the public keys of vehicles or AS, nor can it obtain the complete private keys.

IV. OUR PROPOSED SCHEME

This section details the construction of the proposed certificateless signcryption protocol, which is built upon NTRU lattices and supports batch verification. Key notations are summarized in Table II, followed by a comprehensive description of the NTRU-CLSC scheme.

TABLE II
PARAMETER SYMBOLS AND DEFINITIONS

Symbol	Definition
p	Security parameter
q	Prime modulus (modulus of finite field $\mathbb{Z}_q$)
σ	Discrete Gaussian distribution parameter
α	Target quality parameter of trapdoor
mpk, msk	System public/private key
$\mathbf{B}$	Antrag trapdoor matrix
$\mathbf{h}$	Core component of system public key
ID_i, ID_{AS}	Real identity of vehicle and AS
PID_i, PID_{AS}	Pseudonym of vehicle and AS
d_i, d_{AS}	Partial private key of vehicle and AS
$S_i^{(t)}, S_{AS}^{(t)}$	t -th period secret value of vehicle and AS
$X_{1,i}, X_{2,i}$	Gaussian sampled vectors
r_e	Random parameter in the encryption phase
k	Encryption key
u_i, v_i	Encryption components
g_i	Ciphertext body
g_{agg}	Batched ciphertext body
t_i	Timestamp of signcrypted text
f_i	Anti-replay counter
$H_i (0 \leq i \leq 5)$	hash functions
T_{min}, T_{max}	minimum / maximum period
t	Key update period
T	dynamically generated period time
r_i, y_i, τ_i	signcryption component
$u_{agg}, v_{agg}, \tau_{agg}$	Batched components
$(sk_i^{(t)}, pk_i^{(t)}), (sk_{AS}^{(t)}, pk_{AS}^{(t)})$	t -th period public / private key of vehicle and AS
$\mathcal{A}_1, \mathcal{A}_2$	Type-I / Type-II adversary

A. Setup Phase

In this phase, The KGC executes Algorithm 1 to generate $\mathbf{B} = \begin{pmatrix} a & b \\ M & N \end{pmatrix} \in \mathbb{Z}_q^{2p \times 2p}$ and $\mathbf{h} = b \cdot a^{-1} \in R_q^\times$ (where $a, b \sim D_{\mathbb{Z}^p, \sigma}$, and $r = (\frac{1}{3} \cdot a + \frac{2}{3} \cdot \frac{1}{\alpha})\sqrt{q}$, $R = (\frac{2}{3} \cdot a + \frac{1}{3} \cdot \frac{1}{\alpha})\sqrt{q}$). The KGC selects six hash functions, whose specific definitions are as follows: $H_0 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^p$, $H_1 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^p$, $H_2 : R_q \rightarrow D_{\mathbb{Z}^p, \sigma}$, $H_3 : \mathbb{Z}_q^p \times \{0, 1\}^* \rightarrow D_\lambda : \{v \in \mathbb{Z}_q^* \mid 0 \leq \|v\| \leq \lambda\}$, $H_4 : \{0, 1\}^* \rightarrow \{0, 1\}^l$, $H_5 : \{0, 1\}^* \times \{0, 1\}^* \times R_q \rightarrow \mathbb{Z}_q^p$ (l denotes the length of the vehicle message).

The system public parameters are defined as $params = \{p, q, \mathbf{h}, H_0, H_1, H_2, H_3, H_4, H_5\}$. The primary key $msk = \mathbf{B}$ is securely stored by the KGC, and the primary public key $mpk = \mathbf{h}$.

B. Anonymity Generation Phase

The KGC randomly selects $K_i \in R_q$ and a timestamp t_i , and chooses a random number $\eta_i \in \mathbb{Z}_q^p$. For the generation of the vehicle V_i 's pseudonym PID_i , the KGC constructs its core components through two steps of computation: $PID_{i,1} = K_i \cdot b$, $PID_{i,2} = H_0(ID_i) * H_1(K_i \cdot \mathbf{h} \parallel t_i) \oplus \eta_i$. The pseudonym PID_i is formed by combining the two pseudonym components with the timestamp, i.e., $PID_i = \{PID_{i,1}, PID_{i,2}, t_i\}$. Similarly, the KGC randomly selects $K_{AS} \in R_q$ and a timestamp t_i , and chooses a random number $\eta_{AS} \in \mathbb{Z}_q^p$. For the generation of the pseudonym PID_{AS} of the AS, the KGC constructs its core components through two steps of computation: $PID_{AS,1} = K_{AS} \cdot b$, $PID_{AS,2} = H_0(ID_{AS}) * H_1(K_{AS} \cdot \mathbf{h} \parallel t_i) \oplus \eta_{AS}$. The pseudonym PID_{AS}

Algorithm 3: Initial Public/Secret Key Generation Phase.

Input: Vehicle pseudonym PID_i , system parameters

$params$, primary secret key msk

Output: Initial key pairs $(sk_i^{(0)}, pk_i^{(0)})$

- 1: $d_i = [S_{1,i}, S_{2,i}] \leftarrow \text{GaussianSampler}(\mathbf{B}, \sigma, PID_{i,2})$
- 2: Vehicle verifies: $S_{1,i} + S_{2,i} \cdot \mathbf{h} = PID_{i,2}$ and $\|(S_{1,i}, S_{2,i})\| \leq \sigma\sqrt{2p}$. If verification fails, reject d_i , else proceed.
- 3: Vehicle selects secret value $s_i^{(0)} = \{s_{1,i}^{(0)}, s_{2,i}^{(0)}\} \leftarrow D_{\mathbb{Z}^p, \sigma}$
- 4: $sk_i^{(0)} = \{d_i, s_i^{(0)}\}$
- 5: $P_i^{(0)} \leftarrow s_{1,i}^{(0)} + s_{2,i}^{(0)} \cdot \mathbf{h}$, $\alpha_i \in R_q$
- 6: $pk_i^{(0)} \leftarrow PID_{i,2} + \alpha_i \cdot P_i^{(0)}$
- 7: **return** $(sk_i^{(0)}, pk_i^{(0)})$

is formed by combining the two pseudonym components with the timestamp, i.e., $PID_{AS} = \{PID_{AS,1}, PID_{AS,2}, t_i\}$.

The vehicle and the AS send registration requests using their real identities ID_i and ID_{AS} , respectively. The KGC computes the pseudonyms PID_i (for the vehicle V_i) and PID_{AS} (for the AS), and adds $\{ID_i, PID_i\}$ and $\{ID_{AS}, PID_{AS}\}$ to the database.

C. Initial Public/Secret Key Generation Phase

The KGC generates the partial private key $d_i = [S_{1,i}, S_{2,i}]$ via $\text{GaussianSampler}(\mathbf{B}, \sigma, PID_{i,2})$, where $\mathbf{B}$ denotes the primary private key, σ is the discrete Gaussian sampling error, and $PID_{i,2}$ is the partial pseudonym corresponding to the vehicle V_i . Then, the KGC transmits d_i to the vehicle V_i through a secure channel. Upon receiving d_i , the vehicle V_i verifies the validity of the following two equations: $S_{1,i} + S_{2,i} \cdot \mathbf{h} = PID_{i,2}$ and $\|(S_{1,i}, S_{2,i})\| \leq \sigma\sqrt{2p}$. After successful verification, the vehicle selects a secret value $s_i^{(0)} = \{s_{1,i}^{(0)}, s_{2,i}^{(0)}\}$ from the discrete Gaussian distribution $D_{\mathbb{Z}^p, \sigma}$. It then combines the partial private key d_i with $s_i^{(0)}$ to form the complete private key of the vehicle: $sk_i^{(0)} = \{d_i, s_i^{(0)}\}$. The vehicle V_i computes $P_i^{(0)} = s_{1,i}^{(0)} + s_{2,i}^{(0)} \cdot \mathbf{h}$, randomly selects $\alpha_i \in R_q$, and calculates the public key as $pk_i^{(0)} = PID_{i,2} + \alpha_i \cdot P_i^{(0)}$. Finally, it outputs $(sk_i^{(0)}, pk_i^{(0)})$ as the initial public-private key pair of the vehicle V_i .

Similarly, the KGC obtains the partial private key $d_{AS} = [S_{1,AS}, S_{2,AS}]$ via $\text{GaussianSampler}(\mathbf{B}, \sigma, PID_{AS,2})$, where $PID_{AS,2}$ is the partial pseudonym corresponding to AS. The KGC then sends d_{AS} to AS through a secure channel. Upon receiving d_{AS} , the AS verifies the following two equations: $S_{1,AS} + S_{2,AS} \cdot \mathbf{h} = PID_{AS,2}$ and $\|(S_{1,AS}, S_{2,AS})\| \leq \sigma\sqrt{2p}$. After successful verification, the AS selects a secret value $s_{AS}^{(0)} = \{s_{1,AS}^{(0)}, s_{2,AS}^{(0)}\}$ from $D_{\mathbb{Z}^p, \sigma}$, and combines d_{AS} with $s_{AS}^{(0)}$ to form the complete private key of the AS: $sk_{AS}^{(0)} = \{d_{AS}, s_{AS}^{(0)}\}$. The AS computes $P_{AS}^{(0)} = s_{1,AS}^{(0)} + s_{2,AS}^{(0)} \cdot \mathbf{h}$, randomly selects $\alpha_{AS} \in R_q$, and calculates the public key as $pk_{AS}^{(0)} = PID_{AS,2} + \alpha_{AS} \cdot$

Algorithm 4: Dynamic Secret Key Update Phase.

Input: Current period t , vehicle private key $sk_i^{(t)} = \{d_i, s_i^{(t)}\}$, system parameters $params$

Output: Updated key pairs: $(sk_i^{(t+1)}, pk_i^{(t+1)})$

- 1: Define period bounds:
 $T_{\min} \leftarrow \lceil 2\sigma\sqrt{p} \rceil$, $T_{\max} \leftarrow \min(\frac{q}{2}, 2^p)$
- 2: $T \leftarrow H_5(PID_i, PID_{AS}, t \oplus \sigma) \bmod (T_{\max} - T_{\min}) + T_{\min}$
- 3: $\Delta S_i \leftarrow H_2(s_{1,i}^{(t)} \oplus s_{2,i}^{(t)}) \sim D_{\mathbb{Z}^p, \sigma}$
- 4: $s_{1,i}^{(t+1)} \leftarrow s_{1,i}^{(t)} + \Delta S_i \bmod q$
- 5: $s_{2,i}^{(t+1)} \leftarrow s_{2,i}^{(t)} + \Delta S_i \bmod q$
- 6: $sk_i^{(t+1)} \leftarrow \{d_i, s_i^{(t+1)}\}$
- 7: $P_i^{(t+1)} \leftarrow s_{1,i}^{(t+1)} + s_{2,i}^{(t+1)} \cdot \mathbf{h}$
- 8: $pk_i^{(t+1)} \leftarrow PID_{i,2} + \alpha_i \cdot P_i^{(t+1)}$
- 9: **return** $(sk_i^{(t+1)}, pk_i^{(t+1)})$

$P_{AS}^{(0)}$. Finally, it outputs $(sk_{AS}^{(0)}, pk_{AS}^{(0)})$ as the initial public-private key pair of the AS. Algorithm 3 shows the specific details.

D. Dynamic Secret Key Update Phase

In VANETs, vehicles using fixed keys for extended periods are vulnerable to cumulative attacks. Dynamic key update mitigates this risk by periodically refreshing keys while maintaining update efficiency. We set the minimum period time $T_{\min} = \lceil 2\sigma\sqrt{p} \rceil$, which is linked to (p, σ) to keep the refresh interval sufficiently short under the discrete-Gaussian setting. We set the maximum period time $T_{\max} = \min(\frac{q}{2}, 2^p)$, which is bounded by (p, q) to avoid excessively long update periods and the resulting increase in long-term key-exposure risk. Algorithm 4 illustrates this process.

During each period update, the vehicle V_i computes the period time T for the current period t . The period time T is dynamically generated by combining the node identity and random factors via the hash function: $T = H_5(PID_i, PID_{AS}, t \oplus \sigma) \bmod (T_{\max} - T_{\min}) + T_{\min}$ where t is an incrementing integer ($t = 0, 1, 2, \dots$) that auto-increments by 1 ($t \leftarrow t + 1$) after each key update. In the update process, the vehicle V_i calculates the increment $\Delta S_i = H_2(s_{1,i}^{(t)} \oplus s_{2,i}^{(t)}) \sim D_{\mathbb{Z}^p, \sigma}$, and updates the secret value as: $s_{1,i}^{(t+1)} = s_{1,i}^{(t)} + \Delta S_i \bmod q$, $s_{2,i}^{(t+1)} = s_{2,i}^{(t)} + \Delta S_i \bmod q$. The private key is thus updated to $sk_i^{(t+1)} = \{d_i, s_i^{(t+1)}\}$. Meanwhile, the vehicle V_i computes $P_i^{(t+1)} = s_{1,i}^{(t+1)} + s_{2,i}^{(t+1)} \cdot \mathbf{h}$, and the public key is synchronously updated as: $pk_i^{(t+1)} = PID_{i,2} + \alpha_i \cdot P_i^{(t+1)}$.

Similarly, during each period update, the AS computes the period time T for the current period t via the same hash function: $T = H_5(PID_i, PID_{AS}, t \oplus \sigma) \bmod (T_{\max} - T_{\min}) + T_{\min}$. In the update process, AS calculates the increment $\Delta S_{AS} = H_2(s_{1,AS}^{(t)} \oplus s_{2,AS}^{(t)}) \sim D_{\mathbb{Z}^p, \sigma}$, and updates the secret value as: $s_{1,AS}^{(t+1)} = s_{1,AS}^{(t)} + \Delta S_{AS} \bmod q$, $s_{2,AS}^{(t+1)} = s_{2,AS}^{(t)} + \Delta S_{AS} \bmod q$. The private key is updated to $sk_{AS}^{(t+1)} =$

$\{d_{AS}, s_{AS}^{(t+1)}\}$. Meanwhile, AS computes $P_{AS}^{(t+1)} = s_{1,AS}^{(t+1)} + s_{2,AS}^{(t+1)} \cdot \mathbf{h}$, and the public key is synchronously updated as: $pk_{AS}^{(t+1)} = PID_{AS,2} + \alpha_{AS} \cdot P_{AS}^{(t+1)}$.

E. Signcryption Phase

In this phase, the vehicle V_i performs the signcryption operation. It takes the pseudonym PID_i of the vehicle V_i , the pseudonym PID_{AS} of ID_{AS} , the public-private key pair $(sk_i^{(t)}, pk_i^{(t)})$ of V_i , the public key $pk_{AS}^{(t)}$ of ID_{AS} , the timestamp t_i , and the plaintext m_i as inputs. It outputs the signcrypted text C_i and sends this ciphertext to ID_{AS} . Algorithm 5 illustrates this process.

The vehicle V_i first maintains an incrementing counter $f_i \in \mathbb{Z}$ (initialized to 0) to prevent replay attacks within the same timestamp. To ensure zero-knowledge property and prevent secret key leakage, the vehicle performs a rejection sampling process: It randomly selects $X_{1,i}, X_{2,i} \leftarrow D_{\mathbb{Z}^p, \sigma}$ from the discrete Gaussian distribution. For the signature part, it calculates $r_i = H_3(PID_i, m_i, X_{1,i} + X_{2,i} \cdot \mathbf{h}, t_i)$ to bind the hash value of the message and random parameters, ensuring the uniqueness of the signature. Next, compute:

$$y_i = \begin{pmatrix} y_{1,i} \\ y_{2,i} \end{pmatrix} = \begin{pmatrix} S_{1,i} + \alpha_i \cdot s_{1,i}^{(t)} \\ S_{2,i} + \alpha_i \cdot s_{2,i}^{(t)} \end{pmatrix} \cdot r_i + \begin{pmatrix} X_{1,i} \\ X_{2,i} \end{pmatrix}.$$

Crucially, the vehicle checks whether the euclidean norms satisfy $\|y_{1,i}\| \leq 2\sigma\sqrt{p}$ and $\|y_{2,i}\| \leq 2\sigma\sqrt{p}$. If the condition is not met, the vehicle discards the current $X_{1,i}, X_{2,i}$ and restarts the sampling process. This rejection-sampling step keeps y_i statistically close to a discrete Gaussian distribution and avoids leakage of secret-key information. Under the chosen bound, the acceptance probability is typically a high constant in practice, so only a small expected number of trials is needed. Once satisfied, it forms the signcryption component $\tau_i = (r_i, y_i)$.

The vehicle randomly selects $r_e, e_1, e_2 \leftarrow \{-1, 0, 1\}^p$ and $k \leftarrow \{0, 1\}^p$. Calculate: $u_i = r_e \times \mathbf{h} + \mathbf{e}_1 \in R_q$, $v_i = r_e \times PID_{AS,2} + \mathbf{e}_2 + \lfloor q/2 \rfloor \cdot k \in R_q$, $g_i = m_i \oplus H_4(k)$. Finally, the counter f_i is incremented, and the vehicle outputs the signcrypted text $C_i = (u_i \parallel v_i \parallel g_i \parallel \tau_i \parallel t_i \parallel f_i)$.

F. Unsigncryption Phase

In this phase, AS receives the signcrypted text C_i and performs the unsigncryption operation. The inputs include the pseudonym PID_i of the vehicle V_i , the pseudonym PID_{AS} of ID_{AS} , the private key $sk_{AS}^{(t)}$ of ID_{AS} , the public key $pk_i^{(t)}$ of the vehicle V_i , the timestamp t_i , and the signcrypted text C_i . The output is either the plaintext m_i or $\perp$ (denoting invalidation). Detailed descriptions and the corresponding Algorithm 6 are presented as follows.

First, AS computes the period time T for the current period t , and verifies the freshness of the timestamp t_i . If the timestamp is valid, ID_{AS} calculates its current-period public-private key pair $(sk_{AS}^{(t)}, pk_{AS}^{(t)})$ using the dynamic key update algorithm. Next, it computes: $w_i = v_i - u_i \cdot S_{2,AS}$, $k = \text{round}(\frac{2w_i}{q})$. Then, it decrypts to obtain the message via: $m_i = g_i \oplus H_4(k)$. For

Algorithm 5: Signcryption Phase.

Input: Vehicle pseudonym PID_i , AS pseudonym PID_{AS} , vehicle key pair $(sk_i^{(t)}, pk_i^{(t)})$, AS public key $pk_{AS}^{(t)}$, timestamp t_i , plaintext m_i , System parameters (σ, p)

Output: C_i

- 1: Define Rejection Bound $B \leftarrow 2\sigma\sqrt{p}$
- 2: **repeat**
- 3: Randomly select $X_{1,i}, X_{2,i} \leftarrow D_{\mathbb{Z}^p, \sigma}$
- 4: $r_i \leftarrow H_3(PID_i, m_i, X_{1,i} + X_{2,i} \cdot \mathbf{h}, t_i)$
- 5: $y_i = \begin{pmatrix} y_{1,i} \\ y_{2,i} \end{pmatrix} \leftarrow \begin{pmatrix} S_{1,i} + \alpha_i \cdot s_{1,i}^{(t)} \\ S_{2,i} + \alpha_i \cdot s_{2,i}^{(t)} \end{pmatrix} \cdot r_i + \begin{pmatrix} X_{1,i} \\ X_{2,i} \end{pmatrix}$
- 6: **until** $\|y_{1,i}\| \leq B$ **and** $\|y_{2,i}\| \leq B$
- 7: $\tau_i \leftarrow (r_i, y_i)$
- 8: Randomly select $r_e, e_1, e_2 \leftarrow \{-1, 0, 1\}^p$, $k \leftarrow \{0, 1\}^p$
- 9: $u_i \leftarrow r_e \cdot \mathbf{h} + e_1$
- 10: $v_i \leftarrow r_e \cdot PID_{AS,2} + e_2 + \lfloor q/2 \rfloor \cdot k$
- 11: $g_i \leftarrow m_i \oplus H_4(k)$
- 12: $f_i \leftarrow f_i + 1$
- 13: $C_i \leftarrow (u_i \parallel v_i \parallel g_i \parallel \tau_i \parallel t_i \parallel f_i)$
- 14: **return** C_i

Algorithm 6: Unsigncryption Phase.

Input: Vehicle pseudonym PID_i , AS pseudonym PID_{AS} , AS private key $sk_{AS}^{(t)}$, vehicle public key $pk_i^{(t)}$, timestamp t_i , signcrypted text C_i

Output: m_i or $\perp$

- 1: $T \leftarrow H_5(PID_i, PID_{AS}, t \oplus \sigma)$
mod $(T_{\max} - T_{\min}) + T_{\min}$
- 2: **if** $t_i \notin [\text{current_time} - T, \text{current_time}]$ **then**
- 3: **return** $\perp$
- 4: **end if**
- 5: $(sk_{AS}^{(t)}, pk_{AS}^{(t)}) \leftarrow \text{Algorithm 4}$
- 6: $w_i \leftarrow v_i - u_i \cdot S_{2,AS}$
- 7: $k \leftarrow \text{round}(\frac{2w_i}{q})$
- 8: $m_i \leftarrow g_i \oplus H_4(k)$
- 9: **if** $\|y_{1,i}\| \leq 2\sigma\sqrt{p}$ **and** $\|y_{2,i}\| \leq 2\sigma\sqrt{p}$ **and** $r_i = H_3(PID_i, m_i, y_{1,i} + y_{2,i} \cdot \mathbf{h} - r_i \cdot pk_i^{(t)}, t_i)$ **then**
- 10: **return** m_i
- 11: **else**
- 12: **return** $\perp$
- 13: **end if**

the verification step, AS needs to confirm three conditions: $\|y_{1,i}\| \leq 2\sigma\sqrt{p}$, $\|y_{2,i}\| \leq 2\sigma\sqrt{p}$ (verifying the short vector property of the signature value through norm constraints), and $r_i = H_3(PID_i, m_i, y_{1,i} + y_{2,i} \cdot \mathbf{h} - r_i \cdot pk_i^{(t)}, t_i)$. If all conditions are satisfied, the plaintext m_i is output, otherwise, the signature is deemed invalid, and $\perp$ is output.

G. Data Collection Phase

The RSU, acting as a data collection node, needs to filter invalid signcrypted texts to ensure the reliability of the

collected data. First, it classifies each signcrypted text C_i by its period t . For signcrypted texts in the same period, it first verifies the validity of the timestamp t_i based on the time range corresponding to the period, signcrypted texts C_i that fail the verification are discarded directly. For valid C_i , it further verifies the norm constraint conditions: $\|y_{1,i}\| \leq 2\sigma\sqrt{p}$, $\|y_{2,i}\| \leq 2\sigma\sqrt{p}$. If these constraints are satisfied, the RSU performs the packaging operation: $\tau_{\text{batch}} = \{\tau_1, \tau_2, \tau_3, \dots, \tau_n\}$, $u_{\text{batch}} = \{u_1, u_2, u_3, \dots, u_n\}$, $v_{\text{batch}} = \{v_1, v_2, v_3, \dots, v_n\}$, $g_{\text{batch}} = \{g_1, g_2, g_3, \dots, g_n\}$. Meanwhile, the maximum timestamp $t_{\text{batch}} = \max(t_1, t_2, t_3, \dots, t_n)$ and the counter set $f_{\text{batch}} = \{f_1, f_2, f_3, \dots, f_n\}$ are recorded. Finally, for each distinct period t , the RSU generates the corresponding signcrypted text batch $C_t = (u_{\text{batch}} \parallel v_{\text{batch}} \parallel g_{\text{batch}} \parallel \tau_{\text{batch}} \parallel t \parallel t_{\text{batch}} \parallel f_{\text{batch}})$.

H. Batch Unsigncryption Phase

The AS first retrieves the period t corresponding to the signcrypted text batch C_t , and verifies the uniqueness of the counters f_i in f_{batch} as well as the freshness of t_{batch} . If valid, it computes its public-private key pair $(sk_{AS}^{(t)}, pk_{AS}^{(t)})$ for the current period using the dynamic key update algorithm. For $1 \leq i \leq n$, compute: $w_i = v_i - u_i \cdot S_{2,AS}$, $k = \text{round}(\frac{2w_i}{q})$ then decrypt to obtain the message: $m_i = g_i \oplus H_4(k)$. Next, verify the norm constraint conditions: $\sum_{i=1}^n \|y_{1,i}\| \leq 2n\sigma\sqrt{p}$, $\sum_{i=1}^n \|y_{2,i}\| \leq 2n\sigma\sqrt{p}$. If these constraints are satisfied, further verify whether the following equation holds: $\prod_{i=1}^n r_i = \text{prod}_{i=1}^n H_3(PID_i, m_i, y_{1,i} + y_{2,i} \cdot \mathbf{h} - r_i \cdot pk_i^{(t)}, t_i)$. If all conditions are satisfied, output $\{m_i \mid 1 \leq i \leq n\}$, otherwise, output $\perp$.

I. Correctness Analysis

In this phase, we briefly prove the correctness of the NTRU-CLSC scheme.

• Unsigncryption Verification Analysis

$$\begin{aligned}
& y_{1,i} + y_{2,i} \cdot \mathbf{h} - r_i \cdot pk_i^{(t)} \\
&= \left((S_{1,i} + \alpha_i \cdot s_{1,i}^{(t)}) \cdot r_i + X_{1,i} \right) \\
&\quad + \left((S_{2,i} + \alpha_i \cdot s_{2,i}^{(t)}) \cdot r_i + X_{2,i} \right) \cdot \mathbf{h} \\
&\quad - (PID_{i,2} + \alpha_i \cdot P_i^{(t)}) \cdot r_i \\
&= \left((S_{1,i} + \alpha_i \cdot s_{1,i}^{(t)}) \cdot r_i + X_{1,i} \right) \\
&\quad + \left((S_{2,i} + \alpha_i \cdot s_{2,i}^{(t)}) \cdot r_i + X_{2,i} \right) \cdot \mathbf{h} \\
&\quad - \left((S_{1,i} + S_{2,i} \cdot \mathbf{h}) + \alpha_i \cdot (s_{1,i}^{(t)} + s_{2,i}^{(t)} \cdot \mathbf{h}) \right) \cdot r_i \\
&= X_{1,i} + X_{2,i} \cdot \mathbf{h}
\end{aligned}$$

From this, we can verify the consistency of r_i :

$$\begin{aligned}
r_i &= H_3(PID_i, m_i, X_{1,i} + X_{2,i} \cdot \mathbf{h}, t_i) \\
&= H_3 \left(PID_i, m_i, y_{1,i} + y_{2,i} \cdot \mathbf{h} - r_i \cdot pk_i^{(t)}, t_i \right)
\end{aligned}$$

We derive the term w_i as follows:

$$\begin{aligned}
w_i &= v_i - u_i \cdot S_{2,AS} \\
&= (r_e \times PID_{AS,2} + e_2 + \lfloor q/2 \rfloor \cdot k) \\
&\quad - (r_e \times \mathbf{h} + e_1) \cdot S_{2,AS} \\
&= r_e \times PID_{AS,2} + e_2 + \lfloor q/2 \rfloor \cdot k \\
&\quad - (r_e \times \mathbf{h}) \cdot S_{2,AS} - e_1 \cdot S_{2,AS} \\
&= r_e \times PID_{AS,2} + e_2 + \lfloor q/2 \rfloor \cdot k \\
&\quad - r_e \times (\mathbf{h} \cdot S_{2,AS}) - e_1 \cdot S_{2,AS} \\
&= r_e \times S_{1,AS} + e_2 - e_1 \cdot S_{2,AS} + \lfloor q/2 \rfloor \cdot k
\end{aligned}$$

We then bound the norms of the components:

$$\begin{aligned}
\|r_e \times S_{1,AS}\| &\leq \|r_e\| \cdot \|S_{1,AS}\| \leq \sqrt{p} \cdot \sigma \sqrt{p} = \sigma p \\
\|e_2 - e_1 \cdot S_{2,AS}\| &\leq \sqrt{p} + \sqrt{p} \cdot \sigma \sqrt{p} = \sqrt{p} (1 + \sigma \sqrt{p})
\end{aligned}$$

Combining these bounds:

$$\|r_e \times S_{1,AS} + e_2 - e_1 \cdot S_{2,AS}\| \leq \sigma p + \sqrt{p} (1 + \sigma \sqrt{p})$$

In the scheme, the parameter q is much larger than $\sigma p + \sqrt{p}(1 + \sigma \sqrt{p})$. Thus, the coefficient range of the $\lfloor q/2 \rfloor \cdot k$ term in w_i is $(-q/4, q/4)$, so $k = \text{round}(2w_i/q)$ is consistent with k during signcryption. That is, $m_i = g_i \oplus H_4(k)$ matches the original message m_i . Therefore, the correctness of single unsigncryption for the NTRU-CLSC scheme is proven.

- *Batch Unsigncryption Verification Analysis*

We first verify the norm constraints for terms:

$$\begin{aligned}
\sum_{i=1}^n \|y_{1,i}\| &\leq n \cdot (2\sigma \sqrt{p}) = 2n\sigma \sqrt{p} \\
\sum_{i=1}^n \|y_{2,i}\| &\leq n \cdot (2\sigma \sqrt{p}) = 2n\sigma \sqrt{p}
\end{aligned}$$

Then verify the consistency of the product of r_i :

$$\begin{aligned}
\prod_{i=1}^n r_i &= \prod_{i=1}^n H_3(PID_i, m_i, X_{1,i} + X_{2,i} \cdot \mathbf{h}, t_i) \\
&= \prod_{i=1}^n H_3(PID_i, m_i, y_{1,i} + y_{2,i} \cdot \mathbf{h} - r_i \cdot pk_i^{(t)}, t_i)
\end{aligned}$$

Therefore, the correctness of batch unsigncryption for the NTRU-CLSC scheme is proven.

V. SECURITY ANALYSIS

This section presents a comprehensive security analysis of the proposed NTRU-CLSC scheme tailored for VANET environments.

A. Unforgeability

Theorem 1: If the R -SIS problem is hard for polynomial-time algorithms, then the NTRU-CLSC scheme is unforgeable against Type I adversaries in the random oracle model.

Lemma 1: Assuming the intractability of the $SIS_{q,2,\beta}$ problem (defined with $\beta = \sigma \sqrt{p}(4 + 2\lambda(1 + \lambda))\sqrt{2}$) on the NTRU lattice $\Lambda_{h,q}$, the scheme is secure against forgery attempts by polynomial-time Type I adversaries in the random oracle model.

Proof: Assume there exists a probabilistic polynomial-time (PPT) adversary $\mathcal{A}_1$ capable of forging a valid signcryption tuple with non-negligible advantage. Under this assumption, we can construct a challenger $\mathcal{C}$ that leverages the capabilities of $\mathcal{A}_1$ to solve the underlying NTRU-based SIS instance.

Given a polynomial $\mathbf{h} \in R_q^\times$, the NTRU lattice $\Lambda_{h,q}$, and a real number β , the goal of simulator $\mathcal{C}$ is to output a non-zero vector $(z_1, z_2) \in \Lambda_{h,q}$ such that $0 < \|(z_1, z_2)\| \leq \beta$. $\mathcal{C}$ sets the primary public key $mpk = \mathbf{h}$ and simulates the environment for $\mathcal{A}_1$. A query game is conducted between $\mathcal{A}_1$ and $\mathcal{C}$ as follows.

- *H_0 Oracle Query:* Challenger $\mathcal{C}$ maintains a list L_{H_0} composed of tuples $\{ID_i, PID_{i,2}, H_0(ID_i)\}$. Upon receiving a query for a real identity ID_i , $\mathcal{C}$ scans L_{H_0} to check if an entry for ID_i already exists. If found, $\mathcal{C}$ returns the stored $H_0(ID_i)$. Otherwise, $\mathcal{C}$ randomly selects a vector $H_0(ID_i) \in \mathbb{Z}_q^p$ and a random value $\eta_i \in \mathbb{Z}_q^p$. Then, $\mathcal{C}$ invokes the H_1 oracle simulation to obtain $H_1(K_i \cdot \mathbf{h} \parallel t_i)$ and computes $PID_{i,2} = H_0(ID_i) \cdot H_1(K_i \cdot \mathbf{h} \parallel t_i) \oplus \eta_i$. Finally, $\mathcal{C}$ stores the tuple in L_{H_0} and returns $H_0(ID_i)$ to $\mathcal{A}_1$.
- *H_1 Oracle Query:* Challenger $\mathcal{C}$ maintains a list L_{H_1} storing tuples $\{K_i, t_i, v\}$. When $\mathcal{A}_1$ queries with input (K_i, t_i) , $\mathcal{C}$ checks whether a corresponding tuple exists in L_{H_1} . If it exists, $\mathcal{C}$ returns the stored value v . Otherwise, $\mathcal{C}$ uniformly selects a random vector $v \in \mathbb{Z}_q^p$, records the tuple $\{K_i, t_i, v\}$ in L_{H_1} , and returns v to $\mathcal{A}_1$.
- *H_2 Oracle Query:* Challenger $\mathcal{C}$ maintains a list L_{H_2} of tuples $\{input, \Delta S_i\}$. When $\mathcal{A}_1$ queries with input $S_{1,i}^{(t)} \oplus S_{2,i}^{(t)}$, $\mathcal{C}$ first checks L_{H_2} . If an entry is found, it returns the stored ΔS_i . If not, $\mathcal{C}$ samples $\Delta S_i \leftarrow \mathcal{D}_{\mathbb{Z}^p, \sigma}$ from the discrete Gaussian distribution, stores the new entry in L_{H_2} , and returns ΔS_i to $\mathcal{A}_1$.
- *H_3 Oracle Query:* Challenger $\mathcal{C}$ maintains a list L_{H_3} storing tuples $\{PID_i, m_i, P_{commit}, t_i, r_i\}$. When $\mathcal{A}_1$ inputs $(PID_i, m_i, P_{commit}, t_i)$, $\mathcal{C}$ searches L_{H_3} . If a record is found, the stored r_i is returned. Otherwise, $\mathcal{C}$ randomly selects r_i from the distribution $D_\lambda = \{v \in \mathbb{Z}_q^* \mid 0 \leq \|v\| \leq \lambda\}$. $\mathcal{C}$ then records the tuple in L_{H_3} and returns r_i . Note that P_{commit} corresponds to the value $X_{1,i} + X_{2,i} \cdot \mathbf{h}$.
- *H_4 Oracle Query:* Challenger $\mathcal{C}$ maintains a list L_{H_4} for inputs k . Upon receiving a query for k , $\mathcal{C}$ checks L_{H_4} . If found, it returns the stored hash value. Otherwise, $\mathcal{C}$ uniformly selects a random string from $\{0, 1\}^l$, records it in L_{H_4} , and returns it to $\mathcal{A}_1$.
- *H_5 Oracle Query:* Challenger $\mathcal{C}$ maintains a list L_{H_5} storing tuples $\{PID_i, PID_{AS}, t \oplus \sigma, T\}$. When queried with $(PID_i, PID_{AS}, t \oplus \sigma)$, $\mathcal{C}$ checks L_{H_5} . If the entry exists, it returns the stored period T . Otherwise, $\mathcal{C}$ randomly selects an integer $T \in [T_{min}, T_{max}]$, records the tuple in L_{H_5} , and returns T .
- *User Creation Query:* $\mathcal{C}$ maintains a list L_c to track user keys. When $\mathcal{A}_1$ queries for a user (PID_i, t) , if the entry

exists, $\mathcal{C}$ returns $pk_i^{(t)}$. If not, for the initial period $t = 0$, $\mathcal{C}$ simulates the generation of the partial private key $d_i = [S_{1,i}, S_{2,i}]$ by randomly selecting $S_{1,i}, S_{2,i}$ from the Gaussian distribution and implicitly defining $PID_{i,2} = S_{1,i} + S_{2,i} \cdot \mathbf{h}$. Then, $\mathcal{C}$ selects user secret values $s_i^{(0)}$ and computes $pk_i^{(0)} = PID_{i,2} + \alpha_i \cdot (s_{1,i}^{(0)} + s_{2,i}^{(0)} \cdot \mathbf{h})$. For subsequent periods $t > 0$, $\mathcal{C}$ updates the keys by calling the H_2 oracle simulation. Finally, $\mathcal{C}$ stores the generated keys in L_c and returns $pk_i^{(t)}$.

- **Partial Private Key Query:** When $\mathcal{A}_1$ queries for the partial private key of PID_i , $\mathcal{C}$ retrieves the corresponding $d_i = [S_{1,i}, S_{2,i}]$ from L_c and returns it.
- **Public Key Replacement Query:** When $\mathcal{A}_1$ submits a query with input (PID_i, pk_i') to replace the current public key of user PID_i , challenger $\mathcal{C}$ searches the list L_c for the entry associated with PID_i . If the entry is found, $\mathcal{C}$ replaces the stored public key $pk_i^{(t)}$ with the new value pk_i' and updates the entry in L_c to indicate that the public key has been replaced. This allows $\mathcal{A}_1$ to use the chosen public key pk_i' for subsequent operations involving PID_i .
- **Signcryption Query:** When $\mathcal{A}_1$ requests a signcryption on a message m_i with sender PID_i and receiver PID_{AS} , $\mathcal{C}$ first checks the status of PID_i in L_c . If the public key of PID_i has not been replaced, $\mathcal{C}$ retrieves the valid partial private key d_i and secret value $s_i^{(t)}$ from L_c , and executes the standard Algorithm 5 to generate a valid ciphertext C_i . However, if the public key has been replaced with pk_i' , $\mathcal{C}$ cannot execute the standard algorithm as it lacks the corresponding secret value. In this case, $\mathcal{C}$ performs a simulation: it randomly selects vectors $y_{1,i}, y_{2,i} \in \mathbb{Z}_q^p$ satisfying the norm constraints ($\|y_{1,i}\| \leq 2\sigma\sqrt{p}$, $\|y_{2,i}\| \leq 2\sigma\sqrt{p}$) and a random hash value $r_i \in D_\lambda$. $\mathcal{C}$ then defines the commitment $P_{\text{commit}} = y_{1,i} + y_{2,i} \cdot \mathbf{h} - r_i \cdot pk_i'$ and programs the H_3 oracle such that $H_3(PID_i, m_i, P_{\text{commit}}, t_i) = r_i$. If a collision occurs in H_3 (i.e., this input was previously defined), $\mathcal{C}$ restarts the selection process. After successfully setting r_i , $\mathcal{C}$ sets $\tau_i = (r_i, y_i)$ where $y_i = (y_{1,i}, y_{2,i})^T$. Finally, $\mathcal{C}$ computes the encryption components u_i, v_i, g_i using the receiver's public key pk_{AS} and returns the simulated ciphertext $C_i = (u_i \parallel v_i \parallel g_i \parallel \tau_i \parallel t_i \parallel f_i)$ to $\mathcal{A}_1$. Due to the programmability of the random oracle, this simulated ciphertext is computationally indistinguishable from a real one.
- **Forgery:** Eventually, $\mathcal{A}_1$ halts and outputs a forged ciphertext $C_i^* = (u_i^* \parallel v_i^* \parallel g_i^* \parallel \tau_i^* \parallel t_i^* \parallel f_i^*)$ targeting identity PID^* and message m^* , where $\tau_i^* = (r_i^*, y_i^*)$. For the forgery to be considered valid, two conditions must be met: $\mathcal{A}_1$ has never queried the Signcryption Oracle for (PID^*, m^*) , and $\mathcal{A}_1$ has never queried the Partial Private Key Oracle for PID^* . The forged ciphertext must satisfy the verification equation $r_i^* = H_3(PID^*, m^*, P_{\text{commit}}^*, t_i^*)$, where the commitment is computed as $P_{\text{commit}}^* = y_{1,i}^* + y_{2,i}^* \cdot \mathbf{h} - r_i^* \cdot pk_i^*$.

By applying the Forking Lemma, challenger $\mathcal{C}$ can rewind $\mathcal{A}_1$ to the moment of the H_3 query involving the commitment P_{commit}^*

and provide a distinct random response $r_i' \neq r_i^*$. Consequently, $\mathcal{A}_1$ produces a second valid forgery (r_i', y_i') associated with the same commitment P_{commit}^* . This implies the following equality holds:

$$y_{1,i}' + y_{2,i}' \cdot \mathbf{h} - r_i' \cdot pk_i^* = y_{1,i}^* + y_{2,i}^* \cdot \mathbf{h} - r_i^* \cdot pk_i^*$$

By rearranging the terms, we obtain:

$$(y_{1,i}^* - y_{1,i}') + (y_{2,i}^* - y_{2,i}') \cdot \mathbf{h} - (r_i^* - r_i') \cdot pk_i^* = 0$$

The public key is structured as $pk_i^* = PID_{i,2}^* + \alpha_i^* \cdot P_i^*$, where $PID_{i,2}^* = S_{1,i}^* + S_{2,i}^* \cdot \mathbf{h}$ and $P_i^* = s_{1,i}^{*(t)} + s_{2,i}^{*(t)} \cdot \mathbf{h}$. Substituting these components into the equation, we get:

$$(y_{1,i}^* - y_{1,i}') + (y_{2,i}^* - y_{2,i}') \cdot \mathbf{h} - (r_i^* - r_i') \cdot [(S_{1,i}^* + S_{2,i}^* \cdot \mathbf{h}) + \alpha_i^* \cdot (s_{1,i}^{*(t)} + s_{2,i}^{*(t)} \cdot \mathbf{h})] = 0$$

Grouping the terms by the coefficients of 1 and $\mathbf{h}$, we can extract two vector components z_1 and z_2 in the ring R_q :

$$\begin{aligned} z_1 &= (y_{1,i}^* - y_{1,i}') - (r_i^* - r_i') \cdot (S_{1,i}^* + \alpha_i^* \cdot s_{1,i}^{*(t)}) \\ z_2 &= (y_{2,i}^* - y_{2,i}') - (r_i^* - r_i') \cdot (S_{2,i}^* + \alpha_i^* \cdot s_{2,i}^{*(t)}) \end{aligned}$$

These components satisfy the relation $z_1 + z_2 \cdot \mathbf{h} = 0 \pmod{q}$. Furthermore, based on the properties of the small vectors sampled from the discrete Gaussian distribution and the bounds of the hash outputs, the euclidean norms of z_1 and z_2 are bounded. Specifically:

$$\begin{aligned} \|z_1\| &\leq \|y_{1,i}^*\| + \|y_{1,i}'\| + \|r_i^* - r_i'\| \cdot (\|S_{1,i}^*\| + \|\alpha_i^* s_{1,i}^{*(t)}\|) \\ &\leq 4\sigma\sqrt{p} + 2\lambda \cdot (\sigma\sqrt{2p} + \lambda\sigma\sqrt{p}) \\ &\approx \sigma\sqrt{p}(4 + 2\lambda(1 + \lambda))\sqrt{2} = \beta \end{aligned}$$

The same bound applies to $\|z_2\|$. Since $r_i^* \neq r_i'$, the constructed vector $z = (z_1, z_2)$ is non-zero. Therefore, z constitutes a valid solution to the $\text{SIS}_{q,2,\beta}$ problem over the NTRU lattice $\Lambda_{h,q}$. This contradicts the hardness assumption of the SIS problem. Thus, the NTRU-CLSC scheme is unforgeable against Type I adversaries in the random oracle model. $\square$

Lemma 2: Assuming the intractability of the $\text{SIS}_{q,2,\beta}$ problem (defined with $\beta = \sigma\sqrt{p}(4 + 2\lambda(1 + \lambda))\sqrt{2}$) on the NTRU lattice $\Lambda_{h,q}$, the scheme is secure against forgery attempts by polynomial-time Type II adversaries in the random oracle model.

Proof: Suppose there exists a probabilistic polynomial-time adversary $\mathcal{A}_2$ that can forge a valid signcryption of the NTRU-CLSC scheme with a non-negligible probability. We construct a simulator $\mathcal{C}$ that interacts with $\mathcal{A}_2$ to solve the NTRU-based SIS problem. Given the system parameters $\mathbf{h}$ and the associated lattice $\Lambda_{h,q}$, $\mathcal{C}$ aims to output a non-zero short vector z in $\Lambda_{h,q}$. Unlike the Type I scenario, $\mathcal{A}_2$ models a malicious KGC. Therefore, $\mathcal{C}$ generates the primary secret key $\mathbf{B}$ and the primary public key $\mathbf{h}$ honestly during the setup phase and provides $\mathbf{B}$ to $\mathcal{A}_2$.

It is important to note the specific capabilities of the Type II adversary: since $\mathcal{A}_2$ possesses the primary secret key $\mathbf{B}$, it can

autonomously execute the Gaussian sampling algorithm to generate valid partial private keys d_i for any identity. Consequently, $\mathcal{C}$ does not need to simulate the Partial Private Key Query. Furthermore, by the definition of the Type II security model, $\mathcal{A}_2$ is restricted from replacing public keys to frame honest users. Thus, the Public Key Replacement Query is not available to $\mathcal{A}_2$. The simulation of hash oracles $H_0 \sim H_5$ is identical to that in the proof of Lemma 1. Challenger $\mathcal{C}$ maintains lists $L_{H_0} \sim L_{H_5}$ to ensure consistency. The simulation of the remaining queries proceeds as follows.

- **User Creation Query:** When $\mathcal{A}_2$ issues a query for (PID_i, t) , $\mathcal{C}$ first scans the list L_c . Since $\mathcal{A}_2$ already holds the primary key B and can independently derive d_i , $\mathcal{C}$ focuses on simulating the public key $pk_i^{(t)}$ by distinguishing two scenarios. If $PID_i \neq PID^*$, $\mathcal{C}$ acts honestly, it samples the secret value $s_i^{(t)} \leftarrow D_{\mathbb{Z}^p, \sigma}$, computes the public key $pk_i^{(t)} = PID_{i,2} + \alpha_i \cdot (s_{1,i}^{(t)} + s_{2,i}^{(t)} \cdot \mathbf{h})$, and records the tuple in L_c . If $PID_i = PID^*$, $\mathcal{C}$ embeds the challenge instance by setting $P_i^{(t)} = U^*$ (where $U^* \in R_q$ is a random polynomial) and computing $pk_i^{(t)} = PID_{i,2} + \alpha_i \cdot U^*$. In this case, $\mathcal{C}$ remains ignorant of the secret $s_i^{(t)}$ corresponding to U^* , ensuring that any valid forgery on PID^* essentially reduces to solving the underlying hard problem. Finally, $\mathcal{C}$ returns $pk_i^{(t)}$ to $\mathcal{A}_2$.
 - **Signcryption Query:** When $\mathcal{A}_2$ requests a signcryption on a message m_i with sender PID_i and receiver PID_{AS} , $\mathcal{C}$ distinguishes between two cases based on the sender identity. If $PID_i \neq PID^*$, $\mathcal{C}$ knows the user secret value $s_i^{(t)}$ (as generated in the User Creation Query). Therefore, $\mathcal{C}$ can execute the standard Algorithm 5 honestly using d_i (computed via B) and $s_i^{(t)}$ to generate a valid ciphertext C_i . If $PID_i = PID^*$, $\mathcal{C}$ does not know the secret value $s^{*(t)}$ and must perform a simulation. $\mathcal{C}$ randomly selects vectors $y_{1,i}, y_{2,i} \in \mathbb{Z}_q^p$ satisfying the norm constraints ($\|y_{1,i}\| \leq 2\sigma\sqrt{p}$, $\|y_{2,i}\| \leq 2\sigma\sqrt{p}$) and a random hash value $r_i \in D_\lambda$. $\mathcal{C}$ then calculates the commitment $P_{\text{commit}} = y_{1,i} + y_{2,i} \cdot \mathbf{h} - r_i \cdot pk_i^{(t)}$. To ensure validity, $\mathcal{C}$ programs the H_3 oracle such that $H_3(PID_i, m_i, P_{\text{commit}}, t_i) = r_i$. If a collision occurs in H_3 , $\mathcal{C}$ restarts the process. After setting $\tau_i = (r_i, y_i)$, $\mathcal{C}$ computes the encryption components u_i, v_i, g_i using the receiver's public key pk_{AS} and returns the simulated ciphertext $C_i = (u_i \parallel v_i \parallel g_i \parallel \tau_i \parallel t_i \parallel f_i)$. This output is indistinguishable from a real signcryption.
 - **Forgery:** Eventually, $\mathcal{A}_2$ halts and outputs a forged ciphertext $C_i^* = (u_i^* \parallel v_i^* \parallel g_i^* \parallel \tau_i^* \parallel t_i^* \parallel f_i^*)$ targeting identity PID^* and message m^* , where $\tau_i^* = (r_i^*, y_i^*)$. For the forgery to be considered valid, $\mathcal{A}_2$ must not have queried the Signcryption Oracle for (PID^*, m^*) . The forged ciphertext satisfies the verification equation $r_i^* = H_3(PID^*, m^*, P_{\text{commit}}^*, t_i^*)$, where the commitment is computed as $P_{\text{commit}}^* = y_{1,i}^* + y_{2,i}^* \cdot \mathbf{h} - r_i^* \cdot pk_i^*$.
- By applying the Forking Lemma, challenger $\mathcal{C}$ can rewind $\mathcal{A}_2$ to the moment of the H_3 query involving the commitment P_{commit}^* and provide a distinct random response $r_i' \neq r_i^*$. Consequently,

$\mathcal{A}_2$ produces a second valid forgery (r_i', y_i') associated with the same commitment P_{commit}^* . This implies the following equality holds:

$$y_{1,i}' + y_{2,i}' \cdot \mathbf{h} - r_i' \cdot pk_i^* = y_{1,i}^* + y_{2,i}^* \cdot \mathbf{h} - r_i^* \cdot pk_i^*$$

By rearranging the terms, we obtain:

$$(y_{1,i}' - y_{1,i}^*) + (y_{2,i}' - y_{2,i}^*) \cdot \mathbf{h} - (r_i' - r_i^*) \cdot pk_i^* = 0$$

Recall that the public key is structured as $pk_i^* = PID_{i,2}^* + \alpha_i^* \cdot P_i^*$, where $PID_{i,2}^* = S_{1,i}^* + S_{2,i}^* \cdot \mathbf{h}$ and $P_i^* = s_{1,i}^{*(t)} + s_{2,i}^{*(t)} \cdot \mathbf{h}$. Substituting these components into the equation, we get:

$$(y_{1,i}' - y_{1,i}^*) + (y_{2,i}' - y_{2,i}^*) \cdot \mathbf{h} - (r_i' - r_i^*) \cdot [(S_{1,i}^* + S_{2,i}^* \cdot \mathbf{h}) + \alpha_i^* \cdot (s_{1,i}^{*(t)} + s_{2,i}^{*(t)} \cdot \mathbf{h})] = 0$$

Grouping the terms by the coefficients of 1 and $\mathbf{h}$, we can extract two vector components z_1 and z_2 in the ring R_q :

$$z_1 = (y_{1,i}' - y_{1,i}^*) - (r_i' - r_i^*) \cdot (S_{1,i}^* + \alpha_i^* \cdot s_{1,i}^{*(t)})$$

$$z_2 = (y_{2,i}' - y_{2,i}^*) - (r_i' - r_i^*) \cdot (S_{2,i}^* + \alpha_i^* \cdot s_{2,i}^{*(t)})$$

These components satisfy the relation $z_1 + z_2 \cdot \mathbf{h} = 0 \pmod{q}$. Furthermore, based on the properties of the small vectors sampled from the discrete Gaussian distribution and the bounds of the hash outputs, the euclidean norms of z_1 and z_2 are bounded. Specifically:

$$\begin{aligned} \|z_1\| &\leq \|y_{1,i}'\| + \|y_{1,i}^*\| + \|r_i' - r_i^*\| \cdot (\|S_{1,i}^*\| + \|\alpha_i^* s_{1,i}^{*(t)}\|) \\ &\leq 4\sigma\sqrt{p} + 2\lambda \cdot (\sigma\sqrt{2p} + \lambda\sigma\sqrt{p}) \\ &\approx \sigma\sqrt{p}(4 + 2\lambda(1 + \lambda))\sqrt{2} = \beta \end{aligned}$$

The same bound applies to $\|z_2\|$. Since $r_i^* \neq r_i'$, the constructed vector $z = (z_1, z_2)$ is non-zero. Therefore, z constitutes a valid solution to the $\text{SIS}_{q,2,\beta}$ problem over the NTRU lattice $\Lambda_{h,q}$. This contradicts the hardness assumption of the SIS problem. Thus, the NTRU-CLSC scheme is unforgeable against Type II adversaries in the random oracle model.

B. Confidentiality

Theorem 2: The proposed NTRU-CLSC scheme satisfies IND-CCA2 security against Type I adversaries in the random oracle model, assuming the hardness of the R-LWE problem and the one-wayness of the underlying NTRU encryption.

Proof: We demonstrate the confidentiality of the proposed scheme through a sequence of games, denoted as G_0, G_1 , and G_2 . Let S_i be the event that the adversary $\mathcal{A}_1$ correctly guesses the bit b in Game G_i .

Game 0: This game embodies the standard IND-CCA2 security experiment involving a challenger $\mathcal{C}$ and a Type I adversary $\mathcal{A}_1$. Initially, $\mathcal{C}$ performs the setup algorithm to derive the system parameters and the primary secret key. $\mathcal{A}_1$ receives these parameters and gains access to a suite of oracles. The simulation procedures for hash queries ($H_0 \sim H_5$), as well as queries for User Creation, Partial Private Key Extraction, Public Key Replacement, and Signcryption, mirror those detailed in

the proof of Lemma 1. Regarding the Unsigncryption Query, $\mathcal{C}$ leverages the primary trapdoor $\mathbf{B}$ to reconstruct the receiver's partial private key, thereby facilitating decryption using the full private key (assuming the public key remains original) or following consistent simulation rules. It is explicitly required that $\mathcal{A}_1$ cannot query the partial private key nor replace the public key of the target identity PID_{AS}^* .

Challenge phase: In the challenge phase, $\mathcal{A}_1$ transmits a target identity PID_{AS}^* alongside two distinct messages m_0, m_1 to the challenger. Subsequently, $\mathcal{C}$ selects a random bit $b \in \{0, 1\}$ and constructs the challenge ciphertext $C^* = (u^* \parallel v^* \parallel g^* \parallel \tau^* \parallel t^* \parallel f^*)$ utilizing the receiver's valid public key. During this process, $\mathcal{C}$ derives a session key $k^* \in \{0, 1\}^p$ through the encapsulation mechanism (u^*, v^*) and computes the ciphertext body as $g^* = m_b \oplus H_4(k^*)$. Finally, $\mathcal{A}_1$ returns a conjecture b' . The advantage of $\mathcal{A}_1$ is formulated as

$$\text{Adv}_{\mathcal{A}_1}^{G_0} = |2 \Pr[S_0] - 1|. \quad (2)$$

Game 1: This game deviates from G_0 by altering the simulation of the H_4 oracle specifically within the Challenge Phase. Rather than deriving $g^* = m_b \oplus H_4(k^*)$ via the actual hash of the session key k^* , the challenger $\mathcal{C}$ substitutes the hash output with a uniformly random string $R \in \{0, 1\}^l$, setting $g^* = m_b \oplus R$. To maintain consistency with the random oracle model, $\mathcal{C}$ monitors the H_4 queries made by $\mathcal{A}_1$. If $\mathcal{A}_1$ queries $H_4(k)$ with a value k such that the decryption of the challenge components (u^*, v^*) yields k (i.e., $k = k^*$), we declare that event E has occurred. If event E does not occur, $\mathcal{A}_1$ has no information about the value of $H_4(k^*)$ other than the random string R . Since R is uniformly random, g^* acts as a one-time pad, perfectly hiding the message m_b . Therefore, the difference in the adversary's advantage between G_0 and G_1 is bounded by the probability of event E . Querying k^* without possessing the private key implies that $\mathcal{A}_1$ has successfully inverted the underlying encryption function. Thus,

$$|\Pr[S_0] - \Pr[S_1]| \leq \Pr[E] \leq \text{Adv}_{\text{NTRU}}^{\text{OW-CPA}}(\lambda) \quad (3)$$

Game 2: This game introduces a modification to the generation of the challenge encapsulation pair (u^*, v^*) compared to G_1 . Rather than calculating these components using the standard formulas $u^* = r_e^* \cdot \mathbf{h} + e_1^*$ and $v^* = r_e^* \cdot PID_{AS,2}^* + e_2^* + \lfloor q/2 \rfloor \cdot k^*$, the challenger $\mathcal{C}$ substitutes them with two elements selected uniformly at random from the ring R_q . This alteration is grounded in the Ring Learning With Errors (R-LWE) assumption, which guarantees that samples structured as $(h, r \cdot h + e)$ are computationally indistinguishable from uniform pairs $(h, u) \in R_q \times R_q$. Given that the target public key $PID_{AS,2}^*$ is generated honestly, the valid encryption components are indistinguishable from random values. Consequently, in Game G_2 , (u^*, v^*) reveal no details regarding k^* , and since g^* is blinded by a random string R , the challenge ciphertext C^* becomes statistically independent of the challenge bit b . Thus, the adversary's advantage vanishes in this game. Thus,

$$\Pr[S_2] = 1/2 \quad (4)$$

Combining the analyses of the games, the total advantage of $\mathcal{A}_1$ in the real IND-CCA2 experiment is bounded by:

$$\begin{aligned} \text{Adv}_{\mathcal{A}_1}^{\text{IND-CCA2}}(\lambda) &\leq 2 (\text{Adv}_{\text{NTRU}}^{\text{OW-CPA}}(\lambda) + \text{Adv}_{\text{RLWE}}(\lambda)) \\ &\quad + \text{negl}(\lambda) \end{aligned} \quad (5)$$

Since the R-LWE problem and the one-wayness of NTRU are computationally hard, the advantage is negligible. This concludes the proof for Type I adversaries.

Theorem 3: The proposed NTRU-CLSC scheme satisfies IND-CCA2 security against Type II adversaries in the random oracle model, assuming the hardness of the R-LWE problem and the one-wayness of the underlying NTRU encryption.

Proof: We demonstrate the confidentiality against Type II adversaries through a sequence of games G_0, G_1 , and G_2 . Let S_i be the event that the adversary $\mathcal{A}_2$ correctly guesses the bit b in Game G_i .

Game 0: This game simulates the real IND-CCA2 security experiment with the Type II adversary $\mathcal{A}_2$. The challenger $\mathcal{C}$ executes the setup phase and provides $\mathcal{A}_2$ with the system parameters and the primary secret key $\mathbf{B}$. This models the malicious KGC scenario. $\mathcal{A}_2$ performs oracle queries similar to Game 0 of the Type I proof. However, since $\mathcal{A}_2$ holds $\mathbf{B}$, it can compute initial partial private keys independently. The crucial constraint is that $\mathcal{A}_2$ does not know the user-selected secret value $s_i^{(t)}$ used in the dynamic key update process. Consequently, for the target receiver PID_{AS}^* , $\mathcal{A}_2$ cannot derive the current valid private key $(S_{1,AS}^{(t)}, S_{2,AS}^{(t)})$ for the challenge period t^* , as the update increment ΔS depends on the unknown secret s .

Challenge phase: $\mathcal{A}_2$ submits two messages m_0, m_1 and the target identity. $\mathcal{C}$ computes the challenge ciphertext $C^* = (u^* \parallel v^* \parallel g^* \parallel \tau^* \parallel t^* \parallel f^*)$ using the current public key. $\mathcal{A}_2$ outputs a guess b' . The advantage is

$$\text{Adv}_{\mathcal{A}_2}^{G_0} = |2 \Pr[S_0] - 1|. \quad (6)$$

Game 1: This game is identical to G_0 except for the simulation of the H_4 oracle in the Challenge Phase. $\mathcal{C}$ replaces the hash of the session key $g^* = m_b \oplus H_4(k^*)$ with $g^* = m_b \oplus R$, where $R \in \{0, 1\}^l$ is a truly random string. To maintain consistency, $\mathcal{C}$ monitors H_4 queries. If $\mathcal{A}_2$ queries $H_4(k)$ with the correct session key k^* , event E occurs. Although $\mathcal{A}_2$ knows the primary key, verifying or recovering k^* from the encapsulation (u^*, v^*) without the user's specific secret update information is computationally hard. Thus, if E does not occur, g^* perfectly hides m_b . The difference is bounded by:

$$|\Pr[S_0] - \Pr[S_1]| \leq \Pr[E] \leq \text{Adv}_{\text{NTRU}}^{\text{OW-CPA}}(\lambda) \quad (7)$$

Game 2: This game is identical to G_1 except for the generation of the encapsulation components (u^*, v^*) . Challenger $\mathcal{C}$ replaces u^* and v^* with two uniformly random elements from the ring R_q . Even with knowledge of $\mathbf{B}$, distinguishing the valid encapsulation from random noise relies on the indistinguishability of the underlying R-LWE problem relative to the current period's key structure, which incorporates the user's unknown randomness. In this game, u^* and v^* are random, and g^* is masked by R , so the ciphertext C^* leaks no information about b . The adversary's

advantage is zero. Thus:

$$\Pr[S_2] = 1/2 \quad (8)$$

Combining the analyses, the total advantage of $\mathcal{A}_2$ is bounded by:

$$\text{Adv}_{\mathcal{A}_2}^{\text{IND-CCA2}}(\lambda) \leq 2 (\text{Adv}_{\text{NTRU}}^{\text{OW-CPA}}(\lambda) + \text{Adv}_{\text{RLWE}}(\lambda)) + \text{negl}(\lambda) \quad (9)$$

Since the R-LWE problem and the one-wayness of NTRU are computationally hard, the advantage is negligible. This concludes the proof for Type II adversaries.

C. Security and Privacy Requirements

This subsection examines the security and privacy needs met by our proposed scheme.

- **Resistance to Quantum Attacks:** Our scheme is built upon the NTRU lattice and relies on the hardness of the R-SIS and R-LWE problems, resisting quantum threats without using vulnerable bilinear pairings.
- **Data Integrity:** Integrity is enforced by binding the message to the hash $r_i = H_3(PID_i, m_i, X_{1,i} + X_{2,i} \cdot \mathbf{h}, t_i)$ and checking the norm constraint $\|y_i\| \leq 2\sigma\sqrt{p}$ to detect any tampering.
- **Confidentiality:** Our scheme achieves IND-CCA2 security, ensuring that only the AS possessing the correct private key can decrypt the ciphertext encapsulated via lattice-based operations.
- **User Anonymity:** Vehicles communicate using pseudo-identities (PID_i) generated via discrete Gaussian sampling, ensuring the real identity is never exposed over the public channel.
- **Traceability:** While preserving anonymity, the TA retains the capability to trace the real identity of a malicious vehicle by mapping the pseudo-identity PID_i back to the real registration data.
- **Key Escrow Freedom:** The user's private key $sk_i = [d_i, s_i]$ combines a partial key d_i from KGC and a locally generated secret s_i , preventing the KGC from impersonating users.
- **Resistance to Replay Attacks:** Replay attacks are thwarted by validating the timestamp t_i against the dynamic period $T = H_5(PID_i, PID_{AS}, t \oplus \sigma) \bmod (T_{\max} - T_{\min}) + T_{\min}$.
- **Resistance to Man-in-the-Middle Attacks:** The AS sequentially verifies the signature tuple (r_i, τ_i) and ciphertext integrity, ensuring that any modification to the message or key components during transmission causes verification failure.
- **Resistance to Public Key Replacement:** The receiver verifies the structural binding of the public key to the KGC's primary trapdoor $\mathbf{B}$, preventing Type I adversaries from replacing a user's public key to forge valid signatures.
- **Forward Secrecy:** In each period, the user secret key is refreshed by a new increment derived from the current secret state through H_2 . Therefore, the updated key is not a reuse of the previous-period key. Even if an adversary compromises the current-period secret key, it cannot recover

TABLE III
SECURITY AND PRIVACY COMPARISONS

	Sato et al. [20]	Liu et al. [52]	Zhang et al. [53]	Xu et al. [21]	Jiao et al. [54]	Our Scheme
SPR1	✓	✓	✓	✓	✓	✓
SPR2	✓	✓	✓	✓	✓	✓
SPR3	✓	✓	✓	✓	✓	✓
SPR4	✓	✓	✓	✓	✓	✓
SPR5	×	×	✓	✓	×	✓
SPR6	✓	✓	✓	✓	✓	✓
SPR7	✓	✓	✓	✓	✓	✓
SPR8	✓	✓	✓	✓	✓	✓
SPR9	×	×	✓	×	✓	✓
SPR10	×	×	×	×	×	✓

✓: The requirement is satisfied. ×: The requirement is not satisfied.

previous-period secret keys, and previously signcryptured messages remain protected.

D. Security and Privacy Comparisons

Table III outlines the comparison of these security and privacy requirements against existing quantum-resistant signcryptured schemes [20], [21], [52], [53], [54], where SPR1–SPR10 correspond to Resistance to Quantum Attacks, Data Integrity, Confidentiality, User Anonymity, Traceability, Key Escrow Freedom, Resistance to Replay Attacks, Resistance to MitM Attacks, Resistance to Public Key Replacement, and Forward Secrecy, respectively, and our proposed scheme meets more extensive security and privacy criteria through this analysis, uniquely satisfying resistance to public key replacement and dynamic key updating.

VI. PERFORMANCE ANALYSIS

This section evaluates the performance of the proposed scheme by contrasting it with state-of-the-art quantum-resistant signcryptured protocols [21], [23], [24], [25], [26], [27], specifically focusing on computation and communication costs.

A. Computation Analysis

The experiments are carried out on a PC equipped with Ubuntu 24.04, an i7-11700 CPU, and 16 GB of RAM. We instantiated the proposed scheme with parameters $p = 512$, $q = 12289$ and $\sigma = 2.0$, targeting the NIST security level 1 (128-bit security). The time measurement method involves executing each operation 50 times and calculating the average execution time. Notably, we focus on the most time-consuming operations in the comparison and compare the runtime of key operations across the schemes, as presented in Table IV.

Table V presents a theoretical comparison of the computational costs for the signcryptured and unsigncryptured phases, benchmarking our proposed scheme against six related works [21], [23], [24], [25], [26], [27]. This evaluation is grounded in the operational execution times detailed in Table IV.

TABLE IV
THE EXECUTION TIME OF SEVERAL CRYPTOGRAPHIC OPERATIONS

Operation	Definition	Execution time (ms)
T_{sp}	discrete Gaussian sampling operation	25.3
T_{pis}	SamplePre operation	38.15
T_{htp}	hash operation	1.54×10^{-2}
T_{vmul}	Vector/Polynomial Multiplication operation	17.65
T_{vadd}	Vector/Polynomial Addition operation	4.71×10^{-2}
T_{mod}	Modular Operation	1.9×10^{-3}
T_{norm}	Norm operation	2.8×10^{-3}
T_{xor}	XOR operation	1.9×10^{-3}

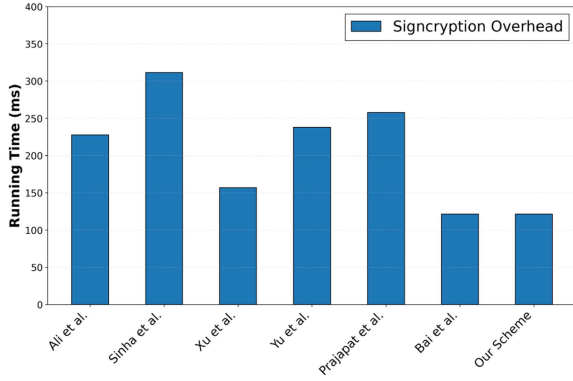


Fig. 2. Running time comparison of signcryption.

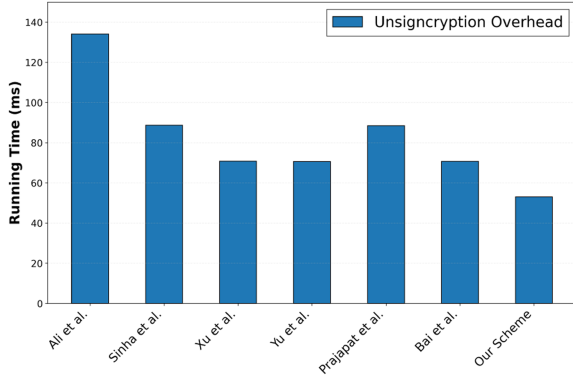


Fig. 3. Running time comparison of unsigncryption.

As illustrated in Table V, the signcryption and unsigncryption overheads of our scheme are 121.42 ms and 53.08 ms, respectively. Figs. 2 and 3 further visualize these comparisons, demonstrating that our scheme maintains a significantly lower time overhead than the referenced protocols.

Fig. 4 presents the comparative analysis of the overall computational overhead. The total overhead encompasses the entire cryptographic lifecycle, including the Setup, Signcryption, and Unsigncryption phases. By aggregating the execution times of the constituent operations, the total computational cost of our scheme is 227.49 ms.

In contrast, the comparative schemes exhibit significantly higher cumulative latencies due to structural inefficiencies. The threshold scheme by Yu et al. [23] and the ring signcryption

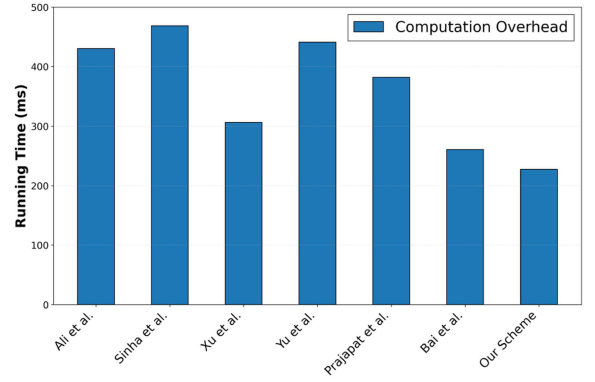


Fig. 4. Comparison of total computation overhead.

scheme by Prajapat et al. [24] suffer from linear scaling costs with respect to the threshold t and ring size M , resulting in total overheads of 441.10 ms and 382.13 ms, respectively. Furthermore, while the Module-LWE based schemes by Xu et al. [21] and Bai et al. [25] provide robust security, they incur heavier setup costs due to the standard TrapGen operation, whereas our scheme utilizes the efficient Antrag trapdoor generation. Ali et al. [26] and Sinha et al. [27] also bring higher cumulative overheads, with total computational costs reaching 430.19 ms and 468.63 ms, respectively. Additionally, the Unsigncryption phase in our scheme is approximately 25% faster than these Module-LWE counterparts due to the optimization of polynomial arithmetic over matrix-vector operations. As evidenced by the data, our proposed scheme reduces the total computational overhead by approximately 12.7% to 51.5% compared to these state-of-the-art solutions. Consequently, our scheme demonstrates superior suitability for delay-sensitive vehicular communication environments.

B. Communication Analysis

We conduct a theoretical analysis of the communication bandwidth required to transmit a signcrypted text. In our evaluation, we set the following parameter sizes to ensure a fair comparison: Let $|m|$ denote the bit length of the message payload (set to 256 bytes), $|Z_q^*|$ denote the bit length of a compressed polynomial element in the lattice ring (set to 512 bytes), and $|t|$ denote the length of the session tag (comprising the timestamp and counter, set to 8 bytes). Additionally, for schemes based on Module-LWE, k represents the module rank, implying that each vector component contains k polynomials, and $|Tag|$ is set to 32 bytes. For ring signcryption, M denotes the number of ring members.

The ciphertext in our proposed NTRU-CLSC scheme is structured as $C_i = (u_i \parallel v_i \parallel g_i \parallel \tau_i \parallel t_i \parallel f_i)$. Specifically, the encapsulation part contains two polynomial components (u_i, v_i) , the encrypted payload g_i corresponds to the message length $|m|$, and the signature component τ_i comprises a hash vector r_i and two signature vectors $y_{1,i}, y_{2,i}$. Consequently, the total communication overhead of our scheme is $|m| + 5|Z_q^*| + |t| = 2824$ bytes.

TABLE V
COMPUTATIONAL OVERHEAD COMPARISON OF DIFFERENT SCHEMES

Schemes	Signcryption Overhead	Unsigncryption Overhead
Xu et al. [21]	$2T_{sp} + 6T_{vmul} + 2T_{htp} + 5T_{vadd} \approx 156.77ms$	$4T_{vmul} + 2T_{htp} + 3T_{vadd} \approx 70.77ms$
Yu et al. [23]	$3T_{pis} + 7T_{vmul} + 2T_{htp} \approx 238.03ms$	$4T_{vmul} + T_{vadd} + T_{htp} \approx 70.66ms$
Prajapat et al. [24]	$6T_{sp} + 6T_{vmul} + T_{htp} \approx 257.72ms$	$5T_{vmul} + 5T_{vadd} + T_{htp} \approx 88.50ms$
Bai et al. [25]	$2T_{sp} + 4T_{vmul} + 2T_{htp} + 4T_{vadd} \approx 121.42ms$	$4T_{vmul} + 2T_{htp} + 2T_{vadd} \approx 70.73ms$
Ali et al. [26]	$5T_{vmul} + 4T_{sp} + 2T_{htp} + T_{pis} \approx 227.63ms$	$4T_{vmul} + T_{sp} + 2T_{htp} + T_{pis} \approx 134.08ms$
Sinha et al. [27]	$6T_{sp} + 9T_{vmul} + 3T_{htp} + 16T_{vadd} + T_{xor} \approx 311.45ms$	$5T_{vmul} + 3T_{htp} + 8T_{vadd} + 6T_{norm} + T_{xor} \approx 88.69ms$
Our Scheme	$2T_{sp} + 4T_{vmul} + 2T_{htp} + 4T_{vadd} \approx 121.42ms$	$3T_{vmul} + 2T_{htp} + 2T_{vadd} + T_{norm} \approx 53.08ms$

TABLE VI
COMMUNICATION OVERHEAD COMPARISON OF DIFFERENT SCHEMES

Scheme	Communication Cost Formula
Xu et al. [21]	$ m + (k + 4) Z_q^* + Tag $
Yu et al. [23]	$ m + 7 Z_q^* + t $
Prajapat et al. [24]	$ m + M \cdot Z_q^* + t $
Bai et al. [25]	$ m + (k + 3) Z_q^* + t $
Ali et al. [26]	$ m + (M + 3) Z_q^* $
Sinha et al. [27]	$ m + (2M + 3) Z_q^* $
Our Scheme	$ m + 5 Z_q^* + t $

Table VI presents a structural comparison with existing quantum-resistant schemes. The threshold scheme by Yu et al. [23] requires a larger ciphertext structure ($7|Z_q^*|$) to accommodate threshold shares and validity proofs, leading to lower transmission efficiency. The three ring signcryption schemes proposed by Prajapat et al. [24], Ali et al. [26], and Sinha et al. [27] all suffer from communication overhead that increases with the ring size M , which results in a noticeable bandwidth burden as the number of participating vehicles grows. Regarding the Module-LWE based schemes by Xu et al. [21] and Bai et al. [25], although their formulas appear constant, they suffer from dimensional expansion. Since they operate on module lattices of rank k , their effective polynomial count is multiplied by k for vector components. For instance, Xu et al. [21] additionally requires a linkability tag $|Tag|$, resulting in a total size of $|m| + (k + 4)|Z_q^*| + |Tag|$, which is significantly larger than our scheme.

In summary, our scheme benefits from the inherent compactness of the NTRU lattice. Unlike the ring-based schemes [24], [26], [27], it avoids communication expansion that scales linearly with the ring size. Unlike Xu et al. [21] and Bai et al. [25], it avoids the dimensional overhead of module lattices. By maintaining a fixed overhead of only $5|Z_q^*| + |t|$, our scheme achieves a better trade-off between security and communication bandwidth, making it highly suitable for bandwidth-constrained vehicular networks.

C. Simulated Analysis

In this subsection, we perform a series of vehicular simulation experiments by leveraging the Veins framework, the intermediate layer of the network simulator OMNeT++, and the traffic flow simulator SUMO, with the aim of verifying the scalability of the proposed protocol in a dynamic, error-prone vehicular network environment. The simulation scenario involves a straight highway segment where the vehicle speed is fixed at 20 m/s, and the vehicle density varies progressively from 10 to 100 vehicles.

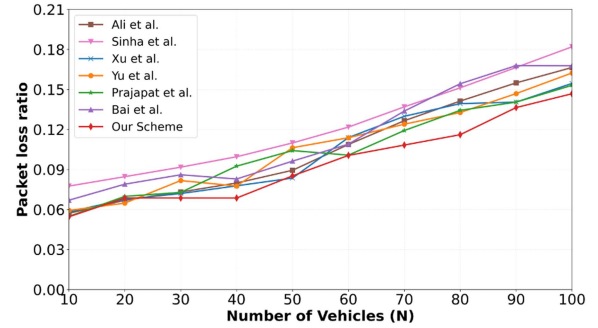


Fig. 5. Impact of vehicle density on packet loss rate for each scheme.

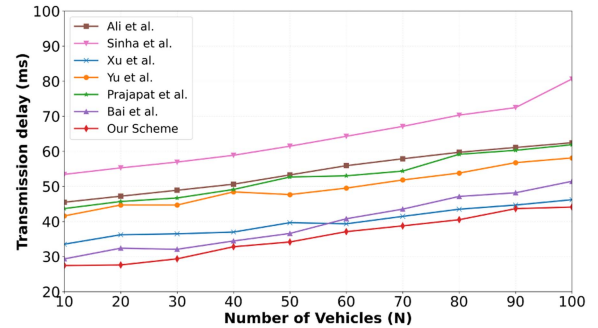


Fig. 6. Impact of vehicle density on transmission delay for each scheme.

We evaluated the variation in transmission delay and packet loss ratio for all comparative schemes.

From Fig. 5, it can be observed that as vehicle density increases, the packet loss ratio of all schemes shows an increasing trend. The proposed scheme has a lower packet loss ratio than those of Bai et al. [25], Yu et al. [23], Ali et al. [26], and Sinha et al. [27], while being slightly superior to those of Prajapat et al. [24] and Xu et al. [21], notably, its growth rate is gentler. Regarding Fig. 6, with the growth of vehicle density, the transmission delay of all schemes generally rises. In contrast, the delay of the proposed scheme remains efficient at 27–44 ms, which is consistently lower than that of all comparative schemes, including Prajapat et al. [24], Yu et al. [23], Ali et al. [26], and Sinha et al. [27]. Although Bai et al. [25] and Xu et al. [21] achieve relatively competitive delay performance under low and medium traffic densities, their delays are still higher than that of our scheme and grow more noticeably as the number of vehicles increases. Although the simulation fixes the vehicle speed at 20 m/s, similar trends are expected at other speeds. Higher speeds may increase topology changes, packet loss, and delay.

However, the cryptographic cost of our scheme is measured per message and is independent of mobility, so its relative efficiency advantage remains unchanged.

In summary, the proposed scheme achieves the best overall performance among all comparative schemes in both packet loss control and transmission delay. It not only maintains lower delay than all competing protocols, but also preserves a comparatively lower packet loss ratio as vehicle density increases. Even when the vehicle density reaches 100, its transmission delay does not exceed 45 ms, and the packet loss ratio stays below 14.7%.

VII. CONCLUSION

In this paper, we proposed an efficient quantum-resistant NTRU lattice-based certificateless signcryption scheme to address the security and efficiency challenges in VANETs. By leveraging the certificateless framework, our design eliminates certificate management burdens and key escrow problems. We provided rigorous formal proofs demonstrating that the scheme satisfies unforgeability and confidentiality under the R-SIS, R-LWE, and NTRU assumptions in the random oracle model. Performance analysis indicates that our scheme significantly outperforms existing lattice-based solutions in both computational efficiency and communication compactness. Simulation results further confirm its superior stability in transmission delay and packet loss control. These analyses demonstrate that the NTRU-CLSC scheme is practical in supporting applications in VANETs environment.

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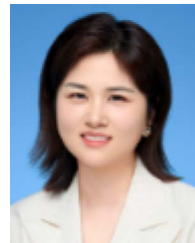
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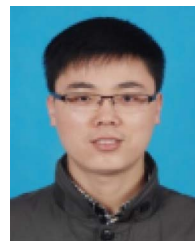
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